

**DEVELOPMENT OF INTELLIGENT ACTIVE FORCE  
CONTROL SCHEME FOR ACCURATE  
TRANSLATIONAL MOVEMENT OF THE END-  
EFFECTOR IN SURGICAL ROBOT MANIPULATOR**

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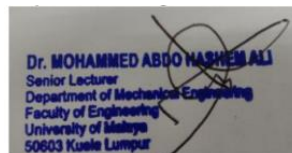
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## ABSTRACT

Robotic-Assisted Surgery (RAS) has emerged as a significant advancement in medical technology, enabling procedures with high precision and minimal invasiveness. However, achieving accurate end-effector translation movement in multi-degree-of-freedom (DOF) manipulators remains challenging due to inherent non-linearities, coupled dynamics, and external disturbances such as surgeon hand tremors. Conventional control strategies, including Proportional-Integral-Derivative (PID) and Computed Torque Control (CTC), often fail to guarantee high-precision tracking under these uncertain dynamic conditions. To address these limitations, this research proposes a robust Intelligent Active Force Control (AFC) scheme integrated with Resolved Acceleration Control (RAC). In this hierarchical architecture, the external RAC loop manages the kinematic trajectory tracking in the Cartesian space, while the inner AFC loop compensates for dynamic disturbances and parameter uncertainties. Crucially, a Fuzzy Logic (FL) mechanism is employed to adaptively estimate the inertia matrix (IN) in real-time, eliminating the reliance on fixed parameters and enhancing disturbance rejection. The system is applied to a 3-DOF RRR (Revolute-Revolute-Revolute) surgical manipulator designed for suturing tasks. The proposed control scheme was validated through rigorous MATLAB/Simulink simulations using a zig-zag suturing trajectory, demonstrating a simulation Root Mean Square Error (RMSE) of 1.28 mm. To evaluate its practical viability, the scheme was implemented and tested on a physical hardware test rig, achieving a hardware validation RMSE of 4.27 mm. In comparative hardware evaluations, the proposed Fuzzy-AFC with RAC scheme outperformed alternative strategies, surpassing a standalone RAC-only controller (which yielded an error of 9.84 mm) and a hybrid Fuzzy-RAC controller (which achieved 12.60 mm). While the system reduced tracking errors compared to the baseline configurations, further optimization is

required to satisfy the strict sub-millimeter accuracy targets demanded for high-precision surgical applications.

## ABSTRAK

Pembedahan Berbantu Robotik (RAS) telah muncul sebagai kemajuan penting dalam teknologi perubatan, membolehkan prosedur dijalankan dengan ketepatan tinggi dan invasif minimum. Walau bagaimanapun, untuk mencapai pergerakan translasi hujung-pengesan (end-effector) yang tepat bagi manipulator pelbagai darjah kebebasan (DOF) kekal mencabar disebabkan oleh ketidaklinearan sedia ada, dinamik gandingan, dan gangguan luaran seperti ketaran tangan pakar bedah. Strategi kawalan konvensional, termasuk Proporsional-Integral-Derifatif (PID) dan Kawalan Tork Terhitung (CTC), sering kali gagal menjamin penjejakan berketepatan tinggi di bawah keadaan dinamik yang tidak menentu ini. Bagi menangani had tersebut, kajian ini mencadangkan satu skema Kawalan Daya Aktif (AFC) Pintar yang teguh bersepadu dengan Kawalan Pecutan Terurai (RAC). Dalam seni bina hierarki ini, gelung luar RAC menguruskan penjejakan trajektori kinematik dalam ruang Cartesian, manakala gelung dalam AFC mengimbangi gangguan dinamik dan ketidakpastian parameter. Secara kritikal, mekanisme Logik Kabur (FL) digunakan untuk menganggar matriks inersia (IN) secara adaptif dalam masa nyata, sekali gus menghapuskan kebergantungan pada parameter tetap dan meningkatkan penolakan gangguan. Sistem ini diaplikasikan pada manipulator pembedahan 3-DOF RRR (Revolute-Revolute-Revolute) yang direka untuk tugas jahitan.

Skema kawalan yang dicadangkan telah disahkan melalui simulasi MATLAB/Simulink yang rapi menggunakan trajektori jahitan zig-zag, mempamerkan Ralat Punca Min Kuasa Dua (RMSE) simulasi sebanyak 2.88 mm. Bagi menilai kebolehlaksanaan praktikalnya, skema ini telah dilaksanakan dan diuji pada pelantar ujian perkakasan fizikal, mencapai RMSE pengesanan perkakasan sebanyak 4.27 mm. Dalam penilaian perkakasan perbandingan, skema Fuzzy-AFC dengan RAC yang dicadangkan mengatasi strategi alternatif, iaitu melepasi pengawal khusus RAC sahaja (yang menghasilkan ralat

sebanyak 9.84 mm) dan pengawal hibrid Fuzzy-RAC (yang mencapai 12.60 mm). Walaupun sistem ini mengurangkan ralat penjejakan berbanding konfigurasi asas, pengoptimuman lanjut masih diperlukan untuk memenuhi sasaran ketepatan bawah-milimeter (sub-millimeter) yang diperlukan bagi aplikasi pembedahan berketepatan tinggi.

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## LIST OF SYMBOLS AND ABBREVIATIONS

|                |                                                           |
|----------------|-----------------------------------------------------------|
| $\hat{\tau}_d$ | estimated torque disturbance                              |
| $\hat{\tau}_d$ | Estimated disturbance in AFC                              |
| $J^\dagger$    | Moore-Penrose Pseudo-Inverse                              |
| $K_I$          | Integral gain                                             |
| $K_d$          | Derivative gain                                           |
| $K_p$          | Proportional gain                                         |
| C              | Coriolis effect                                           |
| E / e          | tracking error                                            |
| $h$            | height of center of mass                                  |
| $J$            | Jacobian                                                  |
| m              | mass                                                      |
| q              | robot's joint configuration vector                        |
| S              | sliding surface, a predefined manifold in the state space |
| sgn            | sign function (signum)                                    |
| T              | Lyapunov candidate function                               |
| $M$            | Moment of Inertia                                         |
| t              | time                                                      |

|          |                                                                                                   |
|----------|---------------------------------------------------------------------------------------------------|
| $\theta$ | angle of rotation for joint in a robotic arm                                                      |
| $l$      | link                                                                                              |
| $r$      | radius of joint to center of mass of link                                                         |
| $I$      | Moment of Inertia                                                                                 |
| $IN$     | inertia matrix                                                                                    |
| $K$      | normalizing output gain/ switching gain                                                           |
| $K$      | Kinetic Energy                                                                                    |
| $L$      | Lagrangian                                                                                        |
| $P$      | Cartesian position of end-effector (DH)                                                           |
| $P$      | Potential Energy                                                                                  |
| $P$      | Symmetric positive-definite matrix (Lyapunov Function)                                            |
| $R$      | end-effector orientation                                                                          |
| $T$      | transformation matrix                                                                             |
| $V$      | scalar, positive-definite function (like energy) of the system's state $x$<br>(Lyapunov function) |
| $a$      | link length (DH)                                                                                  |
| $d$      | link offset (DH)                                                                                  |
| $g/ G$   | gravitational acceleration vector                                                                 |
| $z$      | State vector combining position & velocity error (Lyapunov Function)                              |

|           |                                                                                                                     |
|-----------|---------------------------------------------------------------------------------------------------------------------|
| $\alpha$  | link twist ( $DH$ )                                                                                                 |
| $\alpha$  | Filter factor                                                                                                       |
| $\delta$  | bound of system uncertainties                                                                                       |
| $\eta$    | an arbitrarily small, but positive, constant that defines<br>a neighborhood or a bound around an equilibrium point. |
| $\lambda$ | fractional order of integral term/ sliding surface slope(Fuzzy FOPID)                                               |
| $\mu$     | Fractional order of derivative term (Fuzzy FOPID)                                                                   |
| $\sigma$  | sliding variable or function that defines the desired sliding manifold                                              |
| $\tau$    | torque/ force applied                                                                                               |
| $v$       | Auxiliary input (Sliding Mode Control)                                                                              |
| $v$       | Linear velocity                                                                                                     |
| $\omega$  | angular velocity                                                                                                    |
| AFC       | Active Force Control                                                                                                |
| AFCAIL    | Active Force Control with Iterative Learning                                                                        |
| AI        | Artificial Intelligence                                                                                             |
| CAO       | Chaos-Artemisinin Optimizer                                                                                         |
| CSA       | Clonal Selection Algorithm                                                                                          |
| CTC       | Computed Torque Control                                                                                             |
| DH        | Denavit-Hartenberg                                                                                                  |

|       |                                         |
|-------|-----------------------------------------|
| DOF   | degrees of freedom                      |
| DRL   | Deep Reinforcement Learning             |
| EANN  | Evolutionary Artificial Neural Networks |
| EFJM  | Extended Flexible Joint Model           |
| EP    | Evolutionary Programming                |
| FIS   | Fuzzy Inference System                  |
| FL    | Feedback Linearization                  |
| FL    | Fuzzy Logic                             |
| FLSM  | Feedback Linearization Sliding Mode     |
| FNPN  | Fuzzy Neural Petri Net                  |
| FOPID | Fractional Order PID                    |
| FPGA  | Field-Programmable Gate Array           |
| GA    | Genetic Algorithm                       |
| HILS  | hardware-in-the-loop simulation         |
| IAE   | Integral Absolute Error                 |
| IL    | Iterative Learning                      |
| ISE   | Integral Square Error                   |
| LM    | Levenberg-Marquardt                     |
| LQR   | Linear Quadratic Regulator              |

|        |                                       |
|--------|---------------------------------------|
| LSTM   | Long Short-Term Memory                |
| MEMS   | Micro-Electro-Mechanical Systems      |
| MF     | membership function                   |
| MIMO   | Multi-Input Multi-Output              |
| MIS    | Minimally Invasive Surgery            |
| MLFFNN | Multilayer Feedforward Neural Network |
| MPC    | Model Predictive Control              |
| MSE    | Mean Squared Error                    |
| NN     | Neural Network                        |
| PAM    | pneumatic artificial muscle           |
| PC     | personal computer                     |
| PD     | Proportional-Derivative               |
| PE     | Persistent Excitation                 |
| PID    | Proportional-Integral-Derivative      |
| RAC    | Resolved Acceleration Control         |
| RAS    | Robotic-Assisted Surgery              |
| RBF    | Radial Basis Function                 |
| RMRC   | Resolved Motion Rate Control          |
| RMSE   | Root Mean Square Error                |

|         |                                                                  |
|---------|------------------------------------------------------------------|
| ROISM   | Robust Optimal Integral Sliding Mode Control                     |
| RRR     | Revolute-Revolute-Revolute                                       |
| SCARA   | Selective Compliance Articulated Robot Arm                       |
| SFSMC   | Supervisory Fuzzy Logic Control based Sliding Mode Control       |
| SMC     | Sliding Mode Control                                             |
| ST-FSMC | Self-Tuning Fuzzy Sliding Mode Control                           |
| SVD     | Single Value Decomposition                                       |
|         | Twin Delayed Deep Deterministic Policy Gradient with Prioritized |
| TD3-PER | Experience Reply                                                 |
| VSS     | Variable Structure Systems                                       |
| FLC     | Fuzzy Logic Controller                                           |
| CMR     | Cambridge Medical Robotics                                       |
| CAD     | Computer-aided Design                                            |
| PLA     | Polylactic Acid                                                  |
| SF      | Safety Factor                                                    |
| $A$     | Amplitude                                                        |
| IK      | Inverse Kinematics                                               |

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## CHAPTER 1: INTRODUCTION

### 1.1 Introduction

Robotic manipulators have been widely developed and deployed across various sectors, including medical and surgical applications, due to their ability to enhance precision, consistency, and robustness. Robotic-Assisted Surgery (RAS) represents a major technological advancement aimed at overcoming the limitations of Minimally Invasive Surgery (MIS). Surgical robots enable surgeons to perform procedures with minimal incisions, high precision, and deep access within the human body. Medical robotics also includes tele-operated systems, allowing surgeons to perform procedures remotely. With the advancement of tele-surgery techniques, sensors can capture the surgeon's motion commands outside the operating room and transmit them to a robotic system inside, enabling accurate and responsive surgical manipulation ([Biswas et al., 2023](#)).

Achieving accurate motion and handling of multi-degree-of-freedom robotic manipulators in dynamic and complex environments remains a significant challenge. High-precision trajectory tracking is essential to ensure accurate motion control. However, robotic manipulators are inherently nonlinear, highly coupled Multi-Input Multi-Output (MIMO) systems subject to uncertainties, payload variations, friction, and external disturbances ([Luo, 2024](#)). These characteristics significantly complicate the control problem and make accurate trajectory tracking difficult.

To control the motion of robotic manipulators, conventional strategies such as the Proportional-Integral-Derivative (PID) control and Computed Torque Control (CTC) - also known as Inverse Dynamics Control - are widely used. The PID controller is recognized for its simplicity and flexibility, making it a common choice in industrial automation. However, conventional PID controllers rely on linear assumptions and fixed

gain coefficients, which results in weak adaptability to parameter variations and changing operating conditions. They are generally prone to large tracking errors when applied to complex nonlinear dynamic systems like surgical robots, as they fail to effectively compensate for the highly coupled nature of the system dynamics (Yuan, 2025). Similarly, Computed Torque Control (CTC) utilizes an inner-loop controller to cancel out the robot's nonlinear dynamics mathematically. While this method performs well when dynamic and physical parameters are precisely known, it is highly sensitive to parameter uncertainties (Tilahun, Salau, and Weldcherkos, 2021). In practical surgical scenarios, obtaining a precise mathematical model is often impossible due to varying conditions, making model-based controllers like CTC insufficient for guaranteeing high-precision tracking in the presence of unmodeled dynamics. To address these limitations, robust control strategies are essential.

In this study, a robust controller based on the Active Force Control (AFC) with Resolved Acceleration Control (RAC) for high accurate movement of a robotic manipulator was proposed. Resolved Acceleration Control (RAC) has been identified as a powerful acceleration mode control method that improves upon conventional servo control by handling the kinematic aspects of the manipulator. However, to handle dynamic uncertainties, Active Force Control (AFC) serves as a practical and effective strategy. While conventional AFC relies on a constant estimated mass (Ali and Mailah, 2019), this study employs a Fuzzy Logic mechanism to intelligently estimate the inertia matrix ( $IN$ ) in real-time. Therefore, AFC operates as a superior disturbance rejection scheme that compensates for known and unknown disturbances by adaptively tuning these inertial parameters based on real-time measurements of acceleration and force. Integrating RAC with Fuzzy-AFC allows for a comprehensive control scheme where RAC manipulates the kinematic components while the intelligent AFC compensates for dynamic effects, including bounded disturbances and uncertainties. This combination is

crucial for surgical robots, as it ensures the system remains stable and robust, maintaining precise end-effector positioning even when subjected to vibratory disturbances or changes in payload. The proposed controller was then tested with a robotic manipulator for surgical clinical operations. The proposed controller was validated in simulation and real-time studies and a comparison with existing controllers was conducted.

## 1.2 Problem Statement

The primary problem driving this research is the difficulty of achieving precise end-effector translational movement in surgical manipulators due to internal nonlinearities and external disturbances. Robotic manipulators are multivariable, non-linear, and interconnected dynamic systems. As the degrees of freedom (DOF) increase to enhance flexibility, the complexity of dynamics poses significant challenges for accurate trajectory tracking ([Shatnan and Abd al Arees, 2023](#)). Surgical robots operate in environments where they are subject to various disturbances, such as the high-frequency tremors of a surgeon's hand. These tremors and other external forces can lead to considerable errors in the end-effector's position, compromising safety and precision ([Ghavifekr, et al., 2020](#)). Conventional controllers, such as PID, perform well at low speeds but degrade significantly in the presence of disturbances or when operating at higher speeds. They often lack the ability to effectively compensate for disturbing elements without complex tuning. The performance of AFC relies heavily on the accurate estimation of the robot arm's inertia matrix. Obtaining this estimated inverse dynamic model accurately is difficult, and poor estimation can hinder the controller's ability to decouple actual disturbance torques from applied torques. Therefore, there is a need for a control scheme that can effectively reject disturbances (like tremors) and handle parameter uncertainties while maintaining high kinematic accuracy during surgical procedures.

### 1.3 Research Objectives

1. To design a robust controller based on the combination of Resolve Acceleration Control (RAC) and Fuzzy-based Active Force Control (AFC) for high accurate movement of robotic manipulator.
2. To implement the proposed controller with robotic manipulator for surgical clinical operation.
3. To validate the proposed controller in simulation and real-time studies with comparing to other existing controllers.

### 1.4 Scope of Research

This research focuses on the design, simulation, and physical hardware validation of a hierarchical control framework that pairs Resolved Acceleration Control (RAC) with Fuzzy-Based Active Force Control (AFC) for a surgical robotic manipulator. The investigation is structured around the following parameters:

1. Mathematical Modelling: The dynamic equations mapping joint torques to mechanical motion are established using the energy-based Lagrangian formulation.
2. Control Architecture: A dual-loop system is deployed where the outer RAC loop dictates task-space Cartesian kinematics, while the AFC loop isolates and rejects dynamic disturbances.
3. Intelligent Adaptation: A Mamdani-type Fuzzy Logic Inference System (FIS) utilizes specialized membership functions and error-state rule bases to adaptively estimate the system's diagonal inertia matrix ( $IN$ ) in real-time.

For the trajectory and task boundaries, the control scheme focuses strictly on governing the translational position (X, Y, and Z axes) of the end-effector. Full

orientation control (Roll, Pitch, and Yaw) is completely excluded from this study due to the inherent kinematic limitations of the chosen 3-DOF manipulator model. Trajectory execution is limited to automated, single-plane paths (such as looping, circular, or zigzag patterns) meant to approximate the basic spatial tracking of a surgical needle driver.

Despite the robust design of the integrated controller, the scope of this study is bound by several critical technical and physical limitations:

1. **Tracking Error Threshold:** The control loops are parameterized to restrict the translational tracking error to a boundary within  $\pm 1.0$  mm along the active Cartesian axes.
2. **Payload Capacity:** The manipulator dynamics are structurally optimized for lightweight surgical gear, capping the allowable end-effector payload (representing tissue interaction forces and instrumentation) at 0.5 kg. Exceeding this limit threatens to saturate calculated joint torques and degrade accuracy.
3. **Workspace Envelopes:** The reachable workspace is strictly bounded by the physical link lengths ( $L_1, L_2, L_3$ ) defined in the Denavit-Hartenberg (DH) parameters. Consequently, all automated tasks must be executed within a designated central "sweet spot" to bypass joint limits and avoid kinematic singularities.
4. **Simulation Idealisations:** In the MATLAB/Simulink software environment, prominent physical non-linearities—including joint friction and sensor noise—are not individually modeled; instead, they are globally approximated via a broad mathematical disturbance term inside the AFC loop.
5. **Mechanical Gear Backlash:** On the physical hardware test rig, mechanical gear backlash, joint friction and sensor noise are present within the joint

transmissions, introducing unmodeled non-linear positioning errors during rapid directional adjustments that the controller must actively suppress.

6. Simplified Kinematic Execution: Due to the single-arm configuration and 3-DOF architectural boundaries, the system is fundamentally unable to execute or mimic a full clinical suturing routine, which inherently demands multi-arm tissue tensioning, needle hand-offs, and wound closure physics.
7. Trajectory-Only Imitation: The experimental scope is strictly confined to mimicking the geometric trajectory path of a suturing pattern, omitting physical tissue penetration, needle manipulation forces, and knot-tying operations.

## CHAPTER 2: LITERATURE REVIEW

### 2.1 Introduction

The control of surgical robotic manipulators has evolved through several generations of methodologies, each addressing specific challenges associated with nonlinearity, dynamic coupling, and external disturbances. Early approaches focused on classical feedback control due to their simplicity and ease of implementation. However, as application domains expanded toward high-precision and safety-critical tasks such as surgical manipulation, these methods revealed inherent limitations. Consequently, researchers progressively explored robust, intelligent, and disturbance-rejection-based control strategies. This section critically reviews these approaches, emphasizing their stability characteristics and limitations, and identifies the research gap that motivates the proposed Fuzzy AFC with RAC control scheme.

### 2.2 Surgical Operation-based Medical Robot

The integration of robotics into surgical medicine represents a transformative shift from manual practice to technology-assisted intervention. Recent literature defines surgical robots as intelligent, computer-controlled mechanical systems designed to assist surgeons in planning and performing procedures with enhanced precision, dexterity, and minimal invasiveness ([Dagnino and Kundrat, 2024](#)). These systems are broadly categorized based on their level of autonomy and surgeon interaction, ranging from passive tool-holders to fully teleoperated master-slave systems ([Cepolina and Razzoli, 2022](#)).

While the genesis of surgical robotics is traced back to the PUMA 200 in 1985, recent retrospective reviews emphasize the rapid evolution of these systems into sophisticated multi-functional platforms. [Qian et al. \(2019\)](#) highlight that early systems

like PROBOT and ROBODOC paved the way for modern soft-tissue surgery by demonstrating that robots could surpass human accuracy in rigid translation movements.

The field has since matured into the era of "Digital Surgery" describe the transition from basic endoscopic positioning (e.g., AESOP) to comprehensive telepresence ([Moglia et al., 2016](#)). The current landscape is no longer monopolized by a single system; alongside the established da Vinci Surgical System, newer entrants like the Hugo™ RAS (Medtronic) and Versius (CMR Surgical) have been introduced. These modern systems feature modular arms and open consoles, addressing previous limitations in portability and cost ([Krebs et al., 2023](#); [Peters et al., 2018](#)).

Modern surgical robots predominantly utilize a master-slave teleoperation architecture. In this configuration, the surgeon operates a master console, and the robot (slave) executes movements on the patient. [D'Ettorre et al. \(2018\)](#) note that these systems provide motion scaling (e.g., 5:1 ratio) and tremor filtration, allowing for micro-neurosurgical tasks that were previously impossible manually. As detailed by [Enayati et al. \(2016\)](#), these systems rely heavily on 3D high-definition visualization. However, this reliance on vision has created a dependency on "visual haptics," where surgeons must estimate force based on tissue deformation rather than physical sensation.

Despite the mechanical dexterity of modern systems, significant challenges remain regarding the accurate translation movement of the end-effector, particularly when interacting with soft tissue. A systematic review by [Abiri et al. \(2019\)](#) identifies the lack of force feedback as a critical deficiency. Without haptic cues, surgeons are prone to applying excessive force, leading to tissue trauma or suture breakage during translation movements. Plus, the surgical environment is highly unstructured. [Yang et al. \(2017\)](#) and [Shao et al. \(2025\)](#) discuss how non-linear dynamic forces—such as friction in cable-driven joints and varying tissue stiffness—can degrade positioning accuracy. Standard

Proportional-Integral-Derivative (PID) controllers often fail to compensate for these varying disturbances, resulting in trajectory errors. To address these limitations, recent research has shifted toward Active Force Control (AFC) and impedance control strategies. [He et al. \(2020\)](#) demonstrated that unlike passive compliance, active control schemes can estimate and compensate for external disturbances in real-time.

### 2.3 Conventional Kinematics-based Controllers

Initial investigations into robotic manipulator control relied heavily on conventional PID and PD controllers due to their structural simplicity and minimal computational requirements.

For example, [Kim et al. \(2017\)](#) modeled a three-link (RRR) manipulator using a Proportional Derivative (PD) controller, utilizing a pinhole camera model to generate reference angles from a user's arm motion to guide the robot's trajectory. With relatively low proportional and derivative gains ( $K_p = 0.2$ ,  $K_d = 0.1$ ), the simulation maintained stability; however, it exhibited initial transient overshoot, with steady-state errors eventually contained to less than  $1^\circ$  over a 25-second simulation window.

Similarly, [Kwan, Z.Z. et al. \(2022\)](#) applied a PID controller to a 2-DOF upper limb rehabilitation robot. In repeated motion tests, the system yielded Root Mean Square (RMS) errors of  $2.928 \times 10^{-1}$  for Joint 1 and  $6.287 \times 10^{-1}$  for Joint 2. The rise times were recorded at 0.3712 s and 0.3392 s respectively, demonstrating that while conventional PID can achieve stability, the precision is often limited by fixed gain values.

Although these studies demonstrate that PID-based controllers can maintain system stability under nominal conditions, their performance rapidly degrades in the presence of strong nonlinear coupling, parameter uncertainties, and external disturbances—conditions that are unavoidable in surgical robotic applications. Recent

research has focused on optimizing PID parameters ( $K_p, K_I, K_D$ ) using metaheuristic algorithms and fuzzy logic to enhance trajectory tracking accuracy and stability.

To address the limitations of fixed gains, a Fuzzy Self-Tuning PID controller for a 2-link manipulator was proposed by [Al-khayt \(2012\)](#). The study compared a conventional PID (tuned via Ziegler-Nichols with  $K_p = 45, K_I = 65, K_D = 4$  for a single link test) against the fuzzy variant. The fuzzy supervisor adjusted gains online, achieved a dramatic reduction in maximum overshoot from 3.29% (Conventional PID) to 0.0001% (Fuzzy Self-tuning) and reduced the RMS tracking error from 0.58378 (Conventional PID) to 0.0065 (Fuzzy Self-Tuning). This proves that dynamic gain adjustment significantly mitigates steady-state error in nonlinear systems.

This adaptive capability was further corroborated by [Niu et al. \(2023\)](#) in a cable-driven continuum robot, where a fuzzy inference system using cable length deviation as input significantly decreased overshoot and improved transient response times compared to standard PID.

Beyond online tuning, recent advancements employ complex metaheuristic algorithms to optimize controller coefficients offline. [Shatnan et al. \(2023\)](#) extended PID capabilities by exploring Fractional Order PID (FOPID) controllers optimized via the Clonal Selection Algorithm (CSA). For a 3-DOF robot, the Fuzzy-FOPID controller (with fractional orders  $\lambda = 0.9917, \mu = 0.6292$ ) achieved a Mean Squared Error (MSE) of  $4.308 \times 10^{-7}$ , outperforming standard Fuzzy-PID architectures. This suggests that combining fractional calculus with fuzzy logic provides finer control granularity than integer-order PID.

Concurrently, [Tahiri et al. \(2025\)](#) introduced the Chaos-Artemisinin Optimizer (CAO) to tune a 3-DOF manipulator. The algorithm converged to high-magnitude optimal

gains (e.g.,  $K_p = 516.655$  for joint 1), resulting in an Integral Absolute Error (IAE) of 479.99, significantly outperforming the Cuckoo Search algorithm (IAE = 896.17).

Despite improvements in gain tuning, specific high-precision applications introduce complex dynamics—such as flexibility, haptics, and redundancy—that necessitate specialized control structures. [Morad et al. \(2024\)](#) developed a position control structure for a spine surgery robot with concentric connector joints that integrated active constraints and haptic feedback, successfully reducing encoder signal noise by over 90. Addressing structural elasticity, [Xu et al. \(2023\)](#) proposed a feedforward control method based on an Extended Flexible Joint Model (EFJM) for a 7-DOF robot, reducing position and orientation RMSE by over 60%. Furthermore, to manage kinematic redundancy, [Chang et al. \(2017\)](#) presented a hybrid force/position control algorithm that allowed a 7-DOF manipulator to track paths on arbitrary curved surfaces while maintaining precise force constraints.

In summary, while optimized and intelligent kinematics-based controllers offer incremental improvements over classical closed-loop schemes, their reliance on gain tuning—whether offline or online—limits their robustness when confronted with unmodeled dynamics and unpredictable disturbances. This shortcoming highlights the necessity for more fundamentally robust nonlinear control strategies

## **2.4 Robust and Non-linear Controllers**

To overcome the inherent sensitivity of PID-based controllers to nonlinear dynamics and external disturbances, robust nonlinear control strategies have been extensively investigated. Among these, Sliding Mode Control (SMC) has emerged as a popular solution due to its theoretical robustness and guaranteed stability under bounded uncertainties.

Early implementations, such as those by [Belhocine et al. \(1997\)](#) demonstrated the practical application of Variable Structure Systems (VSS) theory on a SCARA manipulator. To address coupled nonlinear dynamics without relying on velocity sensors, they approximated the system using an identified linear model via the least squares method. Stability was established by forcing the system state to reach and remain on a switching surface  $\sigma(x, t) = 0$ , effectively constraining the system to behave as a stable linear time-invariant system. Experimental results confirmed that derived control gains satisfying the sliding condition  $\sigma\dot{\sigma} < 0$  ensured convergence, limiting position errors to approximately 0.02 rad despite external disturbances.

Building on this foundation, recent research has integrated Feedback Linearization (FL) to handle highly coupled dynamics. [Alqaisi et al. \(2025\)](#) proposed a hybrid Feedback Linearization Sliding Mode (FLSM) controller for a 4-DOF Quanser Qarm. The architecture employed a dual-loop structure: an inner loop utilized input/output feedback linearization to cancel nonlinear dynamics, including the Inertia matrix  $M(\theta)$ , Coriolis/Centrifugal terms  $V(\theta, \dot{\theta})$ , and Gravity vector  $G(\theta)$ , while an outer loop applied an adaptive SMC law to handle bounded uncertainties. The control law utilized a sliding variable  $S = \dot{E} + \Lambda E$ , with an auxiliary input  $v = -K \cdot \text{sgn}(S)$  designed to drive the tracking error  $E$  to zero. Stability was rigorously proved using the Lyapunov candidate function  $V = \frac{1}{2} S^T S$ . By demonstrating that  $\dot{V} \leq -\eta|S|$ , the authors confirmed robustness against uncertainties. Consequently, the FLSM controller reduced the Z-axis Root Mean Square Error (RMSE) from 0.03 (PID) to 0.002, proving its superiority in tracking sharp-edged paths.

Despite these strong theoretical guarantees, conventional SMC suffers from the "chattering" phenomenon—high-frequency switching caused by the signum function—which is physically damaging in surgical applications. To mitigate this, [Qureshi et al.](#)

(2018) developed a Supervisory Fuzzy Logic Control based Sliding Mode Control (SFSMC) for a 2-DOF surgical manipulator. They replaced the discontinuous signum function with a fuzzy inference system (Mamdani type) to smooth the control input and introduced a supervisory layer to tune the output gain  $K$  online based on the tracking error  $e$  and its derivative  $\dot{e}$ . Lyapunov analysis confirmed that as long as the normalizing gain exceeds the bound of system uncertainties ( $K > \delta$ ), the reaching condition  $\dot{V} < 0$  is satisfied. This approach achieved a settling time of 2.8 seconds and a maximum overshoot of only 0.13% (compared to 0.54% for conventional SMC), effectively alleviating chattering.

Similarly, Ashagrie et al. (2021) proposed a Self-Tuning Fuzzy Sliding Mode Controller (ST-FSMC) that replaces the signum function with a saturation function. By utilizing a fuzzy supervisor to online-tune the sliding surface slope  $\lambda$  and switching gain  $K$ , the system achieved a steady-state error of 0.0036 rad while eliminating chattering. Further addressing robustness, Ali et al. (2025) presented a two-stage algorithm combining a Linear Quadratic Regulator (LQR) for payload variations and a Robust Optimal Integral Sliding Mode Control (ROISM) for external disturbances. Phase portrait analysis visually confirmed global asymptotic stability with minimal chattering.

However, while fuzzy and adaptive extensions of SMC successfully mitigate chattering, they introduce significant computational complexity and tuning challenges. This motivates the search for alternative disturbance-rejection mechanisms that are computationally efficient yet robust. Al-Mola and Abdelmaksoud (2025) provided a critical comparison between SMC and Active Force Control (AFC) on a 2-DOF manipulator. While their SMC-PID scheme reduced the Integral Square Error (ISE) from 1.647 (PID) to 0.3797, the AFC-PID scheme achieved a significantly lower ISE of 0.19. This comparative evidence suggests that while SMC is robust, the specific disturbance

rejection mechanism of AFC offers superior performance for handling the dynamic coupling and unmodeled disturbances inherent in surgical robotics, justifying the control strategy proposed in this research.

## 2.5 Artificial Intelligent-based Control System

The effectiveness of the robust nonlinear controllers discussed previously (such as SMC) often hinges on the accuracy of the mathematical model or the precise estimation of physical parameters. This dependency has catalyzed a significant shift toward Artificial Intelligence (AI) based control, where data-driven approaches—ranging from Neural Networks to Deep Reinforcement Learning—are employed to approximate unknown system dynamics without relying on complex, formal mathematical modeling.

Addressing the challenge of unmodeled dynamics, [Luo \(2024\)](#) proposed an adaptive neural network (NN) controller to enhance the grasping accuracy of manipulators. The methodology integrates a Radial Basis Function (RBF) neural network to approximate closed-loop dynamics, combined with a Long Short-Term Memory (LSTM) algorithm. The LSTM structure utilizes input, forget, and output gates specifically designed to mitigate gradient explosion issues during training, thereby refining trajectory correction commands. Stability was rigorously analyzed using Lyapunov theory; the author proved that under Partial Persistent Excitation (PE) conditions, all signals in the closed-loop system remain uniformly bounded, and the tracking error converges asymptotically to zero ( $\lim_{t \rightarrow \infty} e(t) = 0$ ).

In parallel with dynamic control, researchers have applied AI to resolve complex kinematics. [Sharkawy and Khairullah \(2023\)](#) developed a Multilayer Feedforward Neural Network (MLFFNN) trained via the Levenberg-Marquardt (LM) algorithm to resolve both forward and inverse kinematics for a 3-DOF articulated manipulator. By training the network on sinusoidal joint motion data, the system achieved extremely low

Mean Squared Errors (MSE)—approximately  $4.592 \times 10^{-8}$  for forward kinematics and  $9.071 \times 10^{-7}$  for inverse kinematics—indicating that data-driven mapping can effectively bypass the computational burden of analytical Jacobian matrix inversion. Expanding on this hybrid approach, [Zayer et al. \(2021\)](#) proposed a Fuzzy Neural Petri Net (FNPN). Their methodology fused the graphical representation of Petri nets with neural learning and fuzzy reasoning. This architecture achieved an MSE of  $1.0 \times 10^{-5}$  for forward kinematics and  $1.0 \times 10^{-6}$  for inverse kinematics, demonstrating the capability to direct the arm with high precision without explicit kinematic modeling.

Moving beyond supervised learning, [Yuan \(2025\)](#) proposed an adaptive optimization strategy integrating multimodal perception (vision, force, and IMU) with Deep Reinforcement Learning (DRL). The methodology utilizes a Twin Delayed Deep Deterministic Policy Gradient with Prioritized Experience Replay (TD3-PER) algorithm to dynamically optimize admittance parameters ( $K$ ,  $B$ ) and Model Predictive Control (MPC) weights. Unlike previous studies relying on analytical proofs, this system's robustness was evidenced by experimental metrics: it achieved a positioning accuracy of  $\pm 0.19$  mm, reduced force control overshoot to 2.7%, and maintained a 99.3% success rate in unstructured environments, significantly outperforming traditional PID and fuzzy controls.

However, while AI-based controllers offer impressive adaptability, they are not without significant drawbacks. [Sahu et al. \(2021\)](#) in a comprehensive review of robotic manipulation techniques, noted that while optimization algorithms (like Particle Swarm or Differential Evolution) offer near-optimal solutions, they frequently suffer from convergence issues compared to traditional deterministic methods. Furthermore, AI-based systems are characterized by high computational complexity ("black-box" nature), extensive training data requirements, and limited interpretability. Most critically for

surgical applications, guaranteeing closed-loop stability through rigorous analytical proofs remains challenging for deep learning models.

## 2.6 Active Force Controllers

Unlike sliding-mode-based approaches that rely on high-gain switching actions, Active Force Control (AFC) directly targets the root cause of performance degradation by estimating and compensating for dynamic disturbances in real time.

A systematic literature review on the modelling and control of manipulators for inspection and maintenance in challenging environments by [Pistone et al. \(2024\)](#) highlighted that while kinematic control is mature, dynamic control remains a fertile field for research to ensure stability and accuracy in hazardous environments, particularly for hyper-redundant and continuum systems. Their methodology categorized robots into discrete-joint hyper-redundant, continuum, and mobile manipulators, analyzing kinematic and dynamic control strategies across 121 selected documents.

Seminal works by [Ali and Mailah, 2019](#) laid the foundation for AFC as a disturbance-rejection framework capable of maintaining stability and tracking accuracy under severe uncertainties. Early research identified that the effectiveness of AFC relies heavily on the accurate estimation of the inertial parameters. To address this, an intelligent Active Force Control (AFC) strategy that incorporates a Fuzzy Logic (FL) mechanism was introduced to estimate the inertial parameters of a rigid robot arm. In their methodology, the authors developed a hybrid control framework combining Resolved Acceleration Control (RAC), Active Force Control (AFC), Fuzzy Logic (FL), and a Kalman Filter (KF). They are used as a robust path-tracking controller for a non-holonomic wheeled mobile robot operating in a road roundabout environment. The robot path was generated using a laser simulator graph-search technique, while FL estimated the inertia matrix for AFC and KF reduced sensor noise. Simulation results showed

tracking errors below  $10^{-2}$  mm under disturbances, with the RAC-AFC controller achieving maximum errors of only  $2.5 \times 10^{-4}$  mm (x-axis) and  $5 \times 10^{-4}$  mm (y-axis). Experimental testing further demonstrated high accuracy, with maximum tracking errors of approximately  $7 \times 10^{-5}$  m and  $6 \times 10^{-4}$  m in the x and y directions, respectively, confirming strong disturbance rejection and robust navigation performance.

Furthermore, a novel Laser Simulator Logic (LSL) algorithm integrated with an Active Force Control (AFC) scheme to improve the trajectory tracking performance of wheeled mobile robots (WMRs) operating in constrained and noisy environments. was developed by [Ali et al. \(2021\)](#). The methodology involves using LSL to estimate the robot's inertia moment, a critical parameter in AFC, while overcoming the limitations of conventional fuzzy logic when linguistic variables highly overlap. The proposed controller, named PD-AFC-LS-QC, incorporates proportional-derivative control, AFC, LSL-based inertia estimation, and a quick compensation loop to reject disturbances and sensor noise. Simulation and experimental tests on zigzag and highly curved paths demonstrated that the controller achieved more accurate trajectory tracking, superior disturbance rejection, and greater robustness compared with existing control approaches, validating its effectiveness for real-world robotic navigation.

Following on, [Mailah et al. \(2012\)](#) presented the practical implementation of an Active Force Control (AFC) technique incorporating an Iterative Learning (IL) algorithm (AFCAIL) applied to a pneumatic artificial muscle (PAM) actuated two-link robotic arm. The IL algorithm is utilized to intelligently compute the estimated inertia matrix required for the AFC loop, which was cascaded with a PID controller and validated using a hardware-in-the-loop simulation (HILS) configuration. The experimental results verified that the AFCAIL scheme was highly effective and robust, producing rapid responses and

superior trajectory tracking compared to standard PID control by compensating for the inherent nonlinearities and hysteresis of the pneumatic actuators.

More recently, [Al-Mola and Abdelmaksoud \(2025\)](#) proposed a hybrid AFC-PID scheme for a two-degree-of-freedom (2-DOF) redundant robotic manipulator. Their stability results demonstrated that AFC improved the Integral Square Error (ISE) by over 80% compared to standard PID. However, this study largely relied on standard estimation techniques rather than adaptive intelligence.

Collectively, these studies demonstrate that AFC outperforms conventional controllers in disturbance-dominated environments. However, two critical gaps remained. Most AFC implementations are embedded within joint-space control frameworks and lack explicit coordination with kinematic-level acceleration control strategies. This limitation restricts their effectiveness in tasks requiring precise end-effector translation, such as surgical manipulation. While various estimation methods exist, there is a lack of research combining the kinematic precision of RAC with the adaptive capability of Fuzzy-based Inertia Estimation specifically for surgical robotics.

Therefore, this research proposes a Fuzzy AFC with RAC control scheme. This approach leverages RAC for trajectory planning while utilizing Fuzzy Logic to intelligently estimate the inertia matrix in real-time, ensuring robustness against the dynamic uncertainties inherent in surgical environments.

The comparison between the related works in manipulator robot control is illustrated in Table 2.1:

Table 2.1: A Comparison between Related Works' Controllers

| Author                  | Controller                                             | Stability Characteristics                                                                                                     |
|-------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Kim et al., 2017        | PD Controller                                          | Steady state error $< 1^\circ$                                                                                                |
| Kwan, Z.Z. et al., 2022 | PID Controller                                         | Small tracking error in the range of 0.2928 -0.6287                                                                           |
| Al-Khayyt, 2012         | Fuzzy Self-Tuning PID Controller                       | Small tracking error around 0.0065 with reduction in max. overshoot to 0.0065                                                 |
| Niu et al., 2023        | Fuzzy-logic Controller                                 | Decrease in overshoot and faster transient time constant                                                                      |
| Shatnan et al., 2023    | Fuzzy Fractional Order PID Controller (FOPID)          | MSE is around $4.787 \times 10^{-4}$                                                                                          |
| Tahiri et al., 2025     | Feedforward Control                                    | Reduction in position and orientation RMSE by over 65% and 62%                                                                |
| Belhocine et al., 1997  | Sliding Mode Control                                   | Position error around 0.02 rad                                                                                                |
| Alqaisi et al., 2025    | Feedback Linearization with Sliding Mode Control (SMC) | RMSE is around 0.002                                                                                                          |
| Qureshi et al., 2018    | Supervisory Fuzzy Logic Sliding Mode Control (SFSMC)   | Eliminated chattering, reducing max. overshoot to only 0.13% (conventional SMC gains 0.54%) with 2.8 seconds of settling time |
| Ashagrie et al., 2021   | Self-Tuning Sliding Mode Controller (ST-FSMC)          | Eliminated chattering, bringing to steady-state error of 0.0036 rad                                                           |

|                                |                                                                                                                |                                                                                                                         |
|--------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Ali et al., 2025               | Linear Quadratic Regulator (LQR) with Robust Optimal Integral Sliding Mode Control (ROISMIC)                   | Converged to zero with minimal chattering                                                                               |
| Al-Mola and Abdelmaksoud, 2025 | Sliding Mode Control with PID Controller (SMC-PID)<br>Sliding Mode Control with Active Force Control (SMC-AFC) | Integral square error (ISE) is around 0.3797 (SMC-PID), but even lower for SMC-AFC, reaching 0.19.                      |
| Luo, 2024                      | Radial Basis Function (RBF) neural network with LSTM algorithm                                                 | Tracking error converges to zero as time approaches infinity.                                                           |
| Sharkawy and Khairullah, 2023  | Multilayered Feedforward Neural Network (MLFFNN)                                                               | Achieved low MSE around $4.592 \times 10^{-4}$ for forward kinematics and $9.071 \times 10^{-7}$ for inverse kinematics |
| Zayer et al., 2021             | Fuzzy Neural Petri Net (FNPN) Controller                                                                       | MSE is around 0.00001 for forward kinematics and 0.000001 for inverse kinematics                                        |
| Yuan, 2025                     | Deep Reinforcement Learning (DRL)                                                                              | Reduced force control overshoot to 2.7%, and maintained a 99.3% operation success rate                                  |
| Ali and Mailah, 2019           | Active force control (AFC) with Fuzzy Logic (FL)                                                               | Max. track error of only 0.6 mm.                                                                                        |

|                     |                                                               |                                                             |
|---------------------|---------------------------------------------------------------|-------------------------------------------------------------|
| Ali et al. 2021     | Iterative Learning (IL)<br>with Active Force Control<br>(AFC) | Max track error in outdoor path<br>with obstacle is 5.0 mm. |
| Mailah et al., 2012 | Active Force Control with<br>Iterative Learning (AFC-<br>IL)  | Rapid response and superior<br>trajectory tracking          |

## 2.7 Summary of Literature Review

### 2.7.1 Contribution to Sustainable Development Goals (SDGs)

First, it contributes to SDG3: Good Health and Well-being. The primary objective of improving motion control in Robotic-Assisted Surgery (RAS) is to elevate patient safety and clinical outcomes. Advanced control architectures enable high-precision minimally invasive procedures that operate through smaller incisions. By mitigating external disturbances such as a surgeon's high-frequency hand tremors and counteracting non-linear forces that lead to tissue trauma or suture breakage, this line of research directly preserves human health and enhances the quality of specialized surgical interventions.

Next, it targets the SDG 9: Industry, Innovation, and Infrastructure. This research drives engineering innovation by designing a robust, intelligent hierarchical controller that integrates Resolved Acceleration Control (RAC) with Fuzzy-based Active Force Control (AFC). Traditional industrial configurations rely on fixed-gain linear controllers that struggle with unpredictable, coupled dynamics. Introducing real-time, data-driven fuzzy estimation of the inertia matrix represents a significant upgrade to medical manufacturing infrastructure, contributing to the evolution of high-tech "Digital Surgery" platforms. It presents a computationally efficient, structurally robust alternative to the status quo.

Furthermore, it contributes to SDG17: Partnerships for the Goals as well. By developing high-precision surgical robotic technologies demands localized academic and financial mobilization. This project underscores the importance of institutional resource allocation, drawing from the Universiti Malaya Mechanical Engineering Department Final Year Project (FYP) Budget to procure essential physical materials and testing components. These grassroots academic frameworks establish the cross-disciplinary testing pipelines necessary for future multi-sector collaborations between universities, clinical practitioners, and hardware manufacturers to deploy reliable surgical infrastructure globally.

### **2.7.2 Research Gap and Room for Improvement**

While kinematic-level trajectory tracking methods are relatively mature, achieving reliable dynamic control under severe uncertainties remains a critical challenge. Conventional controllers like PID and Computed Torque Control (CTC) degrade rapidly when confronted with highly coupled non-linearities, unmodeled friction, and external surgical disturbances. Advanced robust methods like Sliding Mode Control (SMC) introduce high-frequency "chattering" that is physically dangerous in an operating room, while artificial intelligence and deep learning alternatives are hindered by intense computational workloads, convergence issues, and an inability to analytically guarantee closed-loop stability. Crucially, most existing Active Force Control (AFC) implementations are strictly restricted to joint-space control loops. There is a distinct lack of research combining the spatial trajectory advantages of Cartesian-space Resolved Acceleration Control (RAC) with the real-time adaptive intelligence of Fuzzy-based Inertia Matrix estimation specifically tailored to solve end-effector translation in surgical robotics.

## CHAPTER 3: METHODOLOGY

### 3.1 Introduction

This chapter outlines the systematic approach employed to develop and validate the proposed Intelligent Active Force Control (AFC) scheme for a surgical robot manipulator. The research framework is structured to bridge the gap between theoretical modeling and practical implementation, ensuring that the proposed control strategy effectively addresses the dynamic uncertainties inherent in surgical environments.

To achieve the project objectives, the work is organized into five distinct phases, ranging from the mechanical realization of the system to the final experimental validation. The overall flow of the methodology is summarized in Figure 3.1. The initial phase involves the mechanical design and fabrication of surgical manipulator arm. This stage focuses on designing the geometry of the experimental robot arm, selecting appropriate actuators and rigid link materials to ensure the kinematic structure is suitable for the required surgical workspace and dexterity. The experimental arm is then fabricated for physical surgical operation testing later.

Following the design process, the mathematical foundation of the robot is established. This involves deriving the Forward and Inverse Kinematics for coordinate transformation between task Cartesian space and joint space configuration. Furthermore, Dynamic Modelling is performed (typically using Lagrangian formulations) to understand the forces and torques governing the robot's motion, which is a prerequisite for designing model-based controllers.

Moving on, the control architecture is developed within a simulation environment (e.g., MATLAB). The control strategy integrates Active Force Control (AFC) to compensate for external vibration, with Resolved Acceleration Control (RAC) as the kinematics-based controller. The Fuzzy Logic is designed here to adaptively tune the

inertia estimation, which is changing continuously based on the link configurations during motion.

The validated control algorithms are then translated into Python code and embedded into the hardware controller (Raspberry Pi). The hardware test rig is a 3 DOF robot manipulator with motor encoders and current sensors. Real time experiments are conducted to evaluate the physical response of the manipulator. The study focuses on its ability to track trajectories precisely with the minimal position tracking error.

Finally, the performance of the proposed Intelligent AFC scheme is quantitatively analysed using error metrics such as tracking error. The simulation and physical experimental results are benchmarked against standard RAC and AFC controllers to validate the superior accuracy and disturbance rejection capabilities of the proposed system.

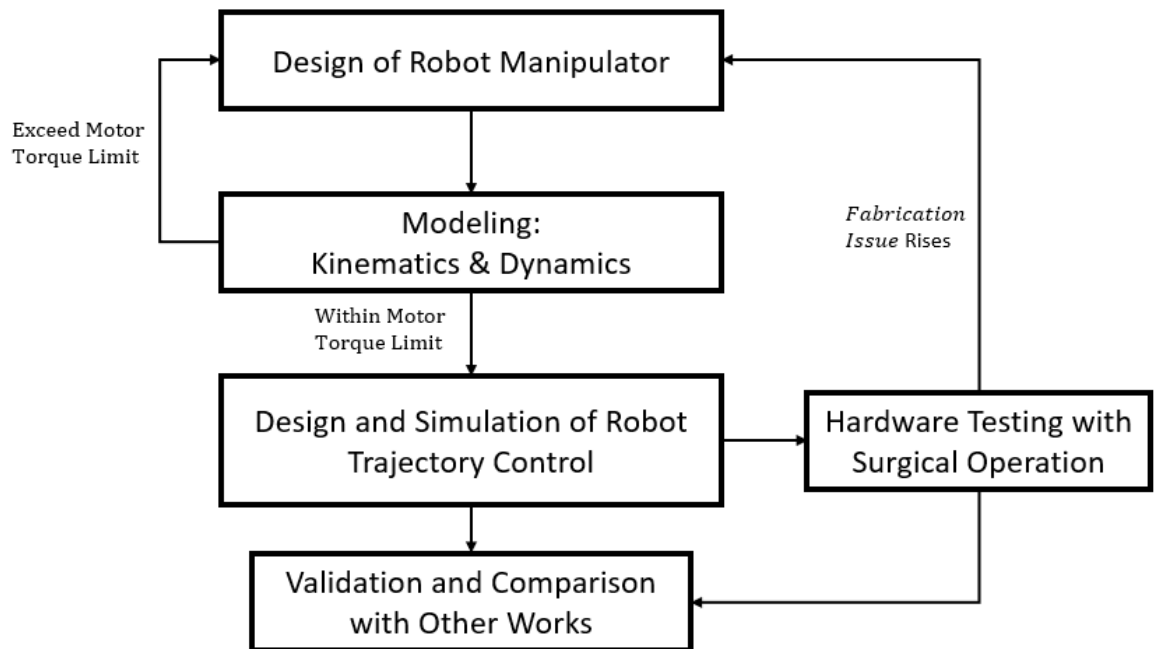


Figure 3.1 Flow Chart

### 3.2 Design and Fabrication of Robot Manipulator

The primary design objective was to develop a robotic arm capable of precise translation movements ( $x$ ,  $y$ ,  $z$ ) to simulate surgical tasks such as suturing. The mechanical structure was first conceptualized and modelled using SolidWorks computer-aided design (CAD) software. the robotic manipulator is modelled as a 3-degree-of-freedom (DOF) serial-chain system composed of rigid links interconnected by revolute joints. The manipulator contains a rotating base for  $z$ -axis rotation and 2 extended links for  $x$ - $y$  axes positioning. The link lengths were determined based on the desired workspace envelope, ensuring the end-effector could reach target coordinates effectively. To minimize the gravitational load on actuators, light-weighted material with high strength-to-weight ratio is selected as the primary fabrication material. To be noted, this will reduce the inertial dynamics that the AFC scheme must compensate for.

The precision of the surgical manipulator is heavily dependent on the performance of its actuators. High torque stepper motor is selected for the joints. To enable closed-loop control, each motor is equipped with a high bit magnetic encoder. These sensors provide real-time feedback on joint angular position and velocity, which are essential inputs for the Active Force Control and Resolved Acceleration Control (RAC) algorithm.

The control system is built around a central processing unit with high processing power. A motor driver is used to amplify the logic signals from the controller to drive the motors. A power supply unit ensures consistent voltage to the actuators, preventing voltage sag during high-load translation movements. Interface boards were used to read encoder data and transmit it to central processing unit for via specific communication protocols.

The links is fabricated and the dimensional accuracy is ensured consistent with the CAD model. This precision is vital, as any discrepancy between the physical link lengths and the theoretical values used in the kinematic equations would introduce

systematic errors in the end-effector position. The assembly process involved mounting the base motor to a fixed platform to eliminate vibrations, coupling the links to the motor shafts using rigid flange coupling to prevent slippage and finally routing the cables through the hollow sections of the links to prevent entanglement during operation.

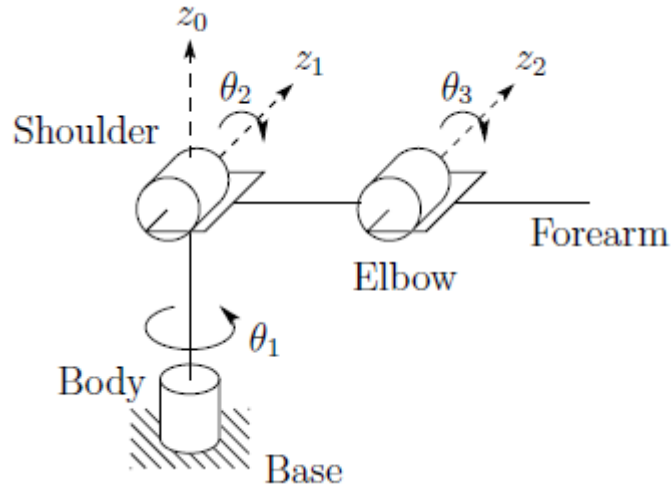


Figure 3.2: 3-DOF RRR Manipulator

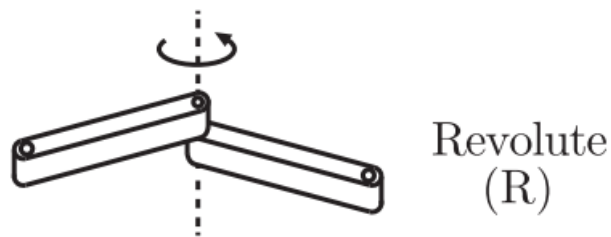


Figure 3.3: Revolute Joints

### 3.2.1 Real World Physical Hardware Testing

#### 3.2.1.1 Structural Fabrication & Backlash Control

The physical links of the 3-DOF RRR manipulator are fabricated using Fused Deposition Modelling (FDM) 3D printing with Polylactic Acid (PLA) filament. PLA provides an advantageous strength-to-weight ratio that yields a low inertial mass, directly easing the dynamic compensation burden on the inner Active Force Control (AFC) loop.

#### 3.2.1.2 Joint Coupling

To maintain a highly rigid mechanical response and avoid the compliance or stretching vulnerabilities typical of belt-driven configurations, the joints utilize direct rigid flange couplings to lock the motor shafts to the links.

#### ***3.2.1.3 Standard Computing and Actuation Architecture***

Unlike distributed multi-stage controller topologies, this hardware setup consolidates the entire processing and execution pipeline into a standalone Raspberry Pi 3B+ single-board computer. The computational framework on the Raspberry Pi 3B+ is responsible for simultaneously executing the high-level Cartesian task-space kinematics loop via Resolved Acceleration Control (RAC), the real-time adaptive Fuzzy Logic Inference System (FIS) to dynamically update the estimated inertia matrix ( $IN$ ), the inner dynamic Active Force Control (AFC) loop for real-time disturbance torque estimation, and direct low-level actuation, generating and delivering the precise pulse-width modulation (PWM), step, and direction command sequences to the stepper motor drivers.

#### ***3.2.1.4 Synchronized Two-Sensor Feedback Instrumentation***

To close the control loops cleanly without minimizing hardware complexity and processing overhead, the experimental rig relies strictly on a synchronized two-sensor configuration per active axis:

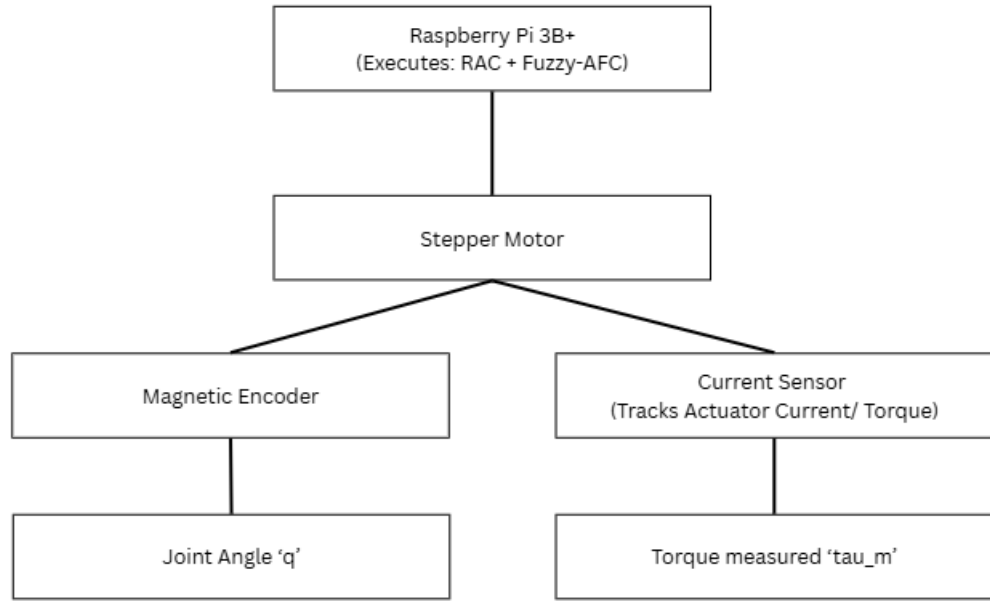


Figure 3.4: Sensors Feedback

High-resolution AS5048A absolute magnetic encoders are mounted directly at the joints to capture real-time angular position data ( $q$ ) at a high precision of approximately  $0.0219^\circ$  (14-bit resolution). These positional values are fed back directly to the outer loop to drive the tracking error vector ( $e$ ) toward zero. Joint velocities and accelerations are computed via numerical differentiation of this clean absolute position signal within the controller environment.

Hall-effect current sensors are placed in line with the joint actuator power stages to track live operational current consumption. Because the output torque of a stepper motor operates as a direct function of its current draw under controlled magnetic excitation, these sensors allow the Raspberry Pi 3B+ to continuously track the actual measured actuator torque ( $\tau_m$ ) in real time. This measured torque is paired with the estimated acceleration inside the inner loop to isolate and calculate the external disturbance torque ( $\hat{\tau}_d$ ) via Equation (58).

### 3.3 Modelling of 3DoF RRR Manipulator

#### 3.3.1 Kinematics of Robotic Arm Manipulator

### 3.3.1.1 Forward Kinematics

Forward Kinematics is the geometric process of calculating the position and orientation of the robot's end-effector (the surgical tool tip) in the Cartesian workspace, given a set of known joint angles ( $q_1, q_2, q_3$ ). For this surgical manipulator, which utilizes a 3-DOF RRR (Revolute-Revolute-Revolute) configuration, the Denavit-Hartenberg (DH) convention is employed to systematically derive these relationships.

To apply the DH convention, coordinate frames are attached to each joint based on the four standard rules:

- Z-axis ( $z_i$ ): This axis must be aligned with the axis of rotation for a revolute joint or the axis of linear translation for a prismatic joint.
- X-axis ( $x_i$ ): This axis must be perpendicular to both the current  $z_i$  axis and the previous  $z_{i-1}$  axis.
- Intersection: The  $x_i$  axis must intersect the previous  $z_{i-1}$  axis (except for the base frame 0).
- Y-axis ( $y_i$ ): This is determined using the Right-Hand Rule based on the established X and Z axes.

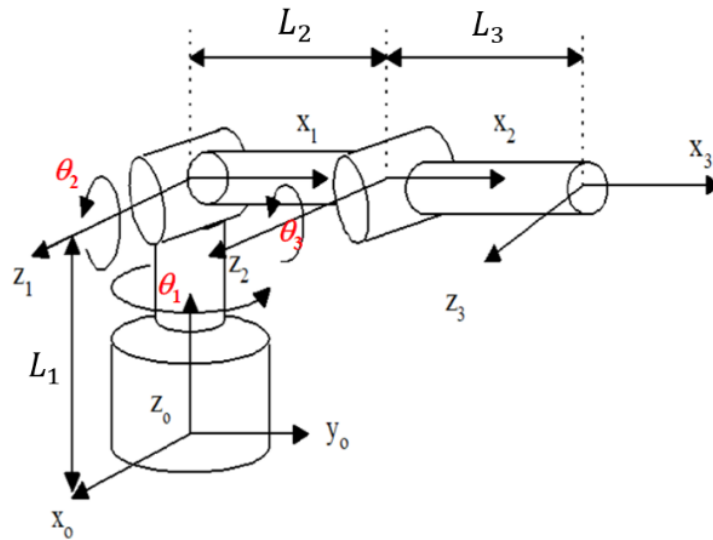


Figure 3.5: Frame Assignment of RRR Manipulator

Based on the frame assignment, the kinematic parameters for the manipulator are identified. The four standard DH parameters are defined as follows:

- $a_i$ : link length (distance along  $x_i$  from  $z_{i-1}$  to  $z_i$ ).
- $\alpha_i$ : link twist (angle about  $x_i$  from  $z_{i-1}$  to  $z_i$ ).
- $d_i$ : link offset (distance along  $z_{i-1}$  from  $x_{i-1}$  to  $x_i$ ).
- $\theta_i$ : joint angle (angle about  $z_{i-1}$  from  $x_{i-1}$  to  $x_i$ ).

For revolute joints,  $\theta_i$  is the joint variable, while  $d_i$  remains constant.

Table 3.1: DH Parameters for Standard RRR manipulator

| $i$ | $a_i$ | $\alpha_i$         | $d_i$ | $\theta_i$ |
|-----|-------|--------------------|-------|------------|
| 1   | 0     | $90^\circ (\pi/2)$ | $L_1$ | $q_1$      |
| 2   | $L_2$ | $0^\circ$          | 0     | $q_2$      |
| 3   | $L_3$ | $0^\circ$          | 0     | $q_3$      |

The relative position and orientation between two adjacent links  $i - 1$  and  $i$  are described by the general homogeneous transformation matrix  $T_{i-1}^i$ :

$$(T_{i-1}^i) = \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_i & \sin\theta_i\sin\alpha_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i\cos\alpha_i & -\cos\theta_i\sin\alpha_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

Substituting the parameters from Table 3.1, the individual transformation matrices are derived:

$$(T_0^1) = \begin{bmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

$$(T_1^2) = \begin{bmatrix} c_2 & -s_2 & 0 & L_2c_2 \\ s_2 & c_2 & 0 & L_2s_2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

$$(T_2^3) = \begin{bmatrix} c_3 & -s_3 & 0 & L_3 c_3 \\ s_3 & c_3 & 0 & L_3 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

where  $c_i = \cos(q_i)$  and  $s_i = \sin(q_i)$ .

The total transformation from the base frame  $\{0\}$  to the end-effector frame  $\{3\}$  is obtained by sequential multiplication of individual link transformation matrices:

$$T_0^3 = T_0^1 \cdot T_1^2 \cdot T_2^3 \quad (5)$$

$$T_0^3 = \begin{bmatrix} c_1 c_{23} & -c_1 s_{23} & s_1 & c_1 (L_2 c_2 + L_3 c_{23}) \\ s_1 c_{23} & -s_1 s_{23} & -c_1 & s_1 (L_2 s_2 + L_3 c_{23}) \\ s_{23} & c_{23} & 0 & L_2 s_2 + L_3 s_{23} + d \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

where  $c_{23} = \cos(q_2 + q_3) = c_2 c_3 - s_2 s_3$  and  $s_{23} = \sin(q_2 + q_3) = c_2 s_3 - s_2 c_3$ .

The matrix follows the general form as below:

$$T_0^n = \begin{bmatrix} R_{0n} & p_{0n} \\ 0_{1 \times 3} & 1 \end{bmatrix} \quad (7)$$

Where

- $R_{0n} \in \mathbb{R}^{3 \times 3}$  represents the end-effector orientation
- $p_{0n} = [x \ y \ z]^T \in \mathbb{R}^3$  denotes the Cartesian position of the end effector.

Given the focus of this research on end-effector translation accuracy, the position vector  $p_{0n}$  is emphasized in controller design and performance evaluation.

### 3.3.1.2 Inverse Kinematics

Inverse Kinematics is the computational process of finding the joint angles for a linked structure, given its end-effector's position and end-effector. Unlike conventional position control schemes that solve for joint angles directly (Geometric Inverse Kinematics), the Resolve Acceleration Control (RAC) scheme operates by commanding

accelerations. Therefore, the inverse kinematics problem is solved at the differential level, which relates Cartesian acceleration to joint acceleration.

The Jacobian matrix  $J(q)$  maps the joint velocities  $\dot{q}$  to the Cartesian velocities  $\dot{x}$  of the end-effector as follows:

$$\dot{x} = J(q)\dot{q} \quad (8)$$

To obtain the relationship required for the RAC scheme, the velocity equation is differentiated with respect to time:

$$\ddot{x} = \frac{d}{dt}(J(q)\dot{q}) \quad (9)$$

$$\ddot{x} = J(q)\ddot{q} + \dot{J}(q)\dot{q} \quad (10)$$

Here,  $\ddot{x}$  represents the Cartesian acceleration. The term  $\dot{J}(q)\dot{q}$  represents the convective acceleration (Coriolis and centrifugal effects reflected in the operational space).

Rearranging the acceleration equation:

$$J(q)\ddot{q}_{ref} = \ddot{x}_{ref} - \dot{J}(q)\dot{q} \quad (11)$$

To solve for  $\ddot{q}_{ref}$ , the Jacobian matrix must be inverted. However, standard matrix inversion is prone to failure if the robot approaches a singular configuration (where  $\det(J) \approx 0$ ). In surgical robotics, singularities can cause dangerous spikes in velocity or torque command. To ensure stability and safety, this study employs the Moore-Penrose Pseudo-Inverse (denoted as  $J^\dagger$ ) rather than the standard inverse. The control law for the inverse kinematics block is formulated as:

$$\ddot{q}_{ref} = J^\dagger(q)(\ddot{x}_{ref} - \dot{J}(q)\dot{q}) \quad (12)$$

Where  $J^f$  is computed using Singular Value Decomposition (SVD). This method provides the minimum-norm solution for joint accelerations, ensuring that the controller remains stable even if the manipulator is fully outstretched or folded (singular poses).

### 3.3.2 Dynamics of Robotics Arm Manipulator

Dynamic modelling provides a mathematical description of the relationship between applied joint actuator torques and the resulting joint motion of the robotic manipulator. Lagrange formulation is employed. Compared to the Newton-Euler method, which relies on vector mechanics and free-body diagrams for every link, the Lagrangian approach is energy-based, without the need to include workless forces and constraint forces to derive the closed-form dynamic equations.

First, we define the 3-DOF robotics arm with the following generalized coordinates (joint angles) as shown in Figure 3.6:

- $\theta_1$ (Waist): Rotation about the vertical Z-axis (Base)
- $\theta_2$ (Shoulder): Rotation about the horizontal axis, perpendicular to Link 1.
- $\theta_3$ (Elbow): Rotation about the horizontal axis, parallel to the Shoulder axis (relative to Link 2)

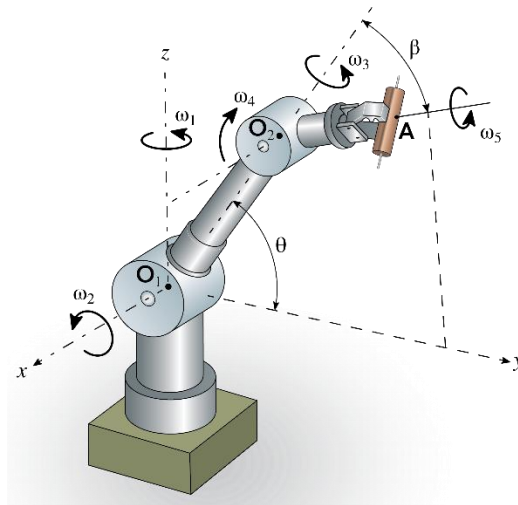


Figure 3.6: Generalized Coordinates for Dynamics

For Link 1 (Base), the linear velocity is  $v_1 = 0$ , while angular velocity is given by:

$$\omega_1 = \dot{\theta}_1 \hat{k} \quad (13)$$

The coordinates of the COM for Link 2 ( $x_2, y_2, z_2$ ) projecting  $r_2$  into the 3D space:

$$x_2 = r_2 \cos \theta_2 \cos \theta_1 \quad (14)$$

$$y_2 = r_2 \cos \theta_2 \sin \theta_1 \quad (15)$$

$$z_2 = r_2 \sin \theta_2 \quad (16)$$

Differentiating with respect to time to find velocities ( $\dot{x}_2, \dot{y}_2, \dot{z}_2$ ):

$$\dot{x}_2 = -r_2 s_2 c_1 \dot{\theta}_2 - r_2 c_2 s_1 \dot{\theta}_1 \quad (17)$$

$$\dot{y}_2 = -r_2 s_2 s_1 \dot{\theta}_2 + r_2 c_2 c_1 \dot{\theta}_1 \quad (18)$$

$$\dot{z}_2 = r_2 c_2 \dot{\theta}_2 \quad (19)$$

Notation:  $s_1 = \sin \theta_1, c_2 = \cos \theta_2$ , etc.

Squared velocity magnitude  $v_2^2 = r_2^2 \cos^2 \theta_2 \dot{\theta}_1^2 + r_2^2 \dot{\theta}_2^2$ :

$$v_2^2 = r_2^2 \cos^2 \theta_2 \dot{\theta}_1^2 + r_2^2 \dot{\theta}_2^2 \quad (20)$$

The coordinates of the COM for Link 3 ( $x_3, y_3, z_3$ ) are given as:

$$x_3 = (l_2 c_2 + r_3 c_{23}) c_1 \quad (21)$$

$$y_3 = (l_2 c_2 + r_3 c_{23}) s_1 \quad (22)$$

$$z_3 = l_2 s_2 + r_3 s_{23} \quad (23)$$

Squared velocity magnitude  $v_3^2$  (after derivation and simplification):

$$v_3^2 = (l_2 c_2 + r_3 c_{23})^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + r_3^2 (\dot{\theta}_2 + \dot{\theta}_3)^2 + 2l_2 r_3 c_3 (\dot{\theta}_2^2 + \dot{\theta}_2 \dot{\theta}_3) \quad (24)$$

The total kinetic energy is the sum of translational and rotational kinetic energy for all links:

$$K = K_1 + K_2 + K_3 \quad (25)$$

$$K_1 = \frac{1}{2} I_{zz1} \dot{\theta}_1^2 \quad (26)$$

$$K_2 = \frac{1}{2} m_2 v_2^2 + \frac{1}{2} I_2 \dot{\theta}_2^2 \quad (27)$$

$$K_2 = \frac{1}{2} m_2 \left( r_2^2 c_2^2 \dot{\theta}_1^2 + r_2^2 \dot{\theta}_2^2 \right) + \frac{1}{2} I_2 \dot{\theta}_2^2 \quad (28)$$

$$K_3 = \frac{1}{2} m_3 v_3^2 + \frac{1}{2} I_3 (\dot{\theta}_2 + \dot{\theta}_3)^2 \quad (29)$$

$$K_3 = \frac{1}{2} m_3 \left[ (l_2 c_2 + r_3 c_{23})^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + r_3^2 (\dot{\theta}_2 + \dot{\theta}_3)^2 + 2 l_2 r_3 c_3 (\dot{\theta}_2^2 + \dot{\theta}_2 \dot{\theta}_3) \right] \quad (30)$$

Potential energy is determined by the height (z) of the center of mass in the gravitational field (g).

$$P_1 = 0 \quad (31)$$

$$P_2 = m_2 g z_2 = m_2 g r_2 \sin \theta_2 \quad (32)$$

$$P_3 = m_3 g z_3 = m_3 g l_2 \sin[\theta_2 + r_3 \sin(\theta_2 + \theta_3)] \quad (33)$$

Total potential energy is given as:

$$P = g[m_2 r_2 s_2 + m_3 (l_2 s_2 + r_3 s_{23})] \quad (34)$$

The Lagrangian is defined as the difference between Kinetic and Potential energy:

$$L = K - P \quad (35)$$

We apply the Euler-Lagrange equation for each joint  $i=1, 2, 3$ :

$$\tau_i = \frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}_i} \right) - \frac{\partial L}{\partial \theta_i} \quad (36)$$

where  $\tau_i$  represents the generalized torque applied at joint  $i$ .

For joint 1  $\theta_1$  motion, since P does not depend on  $\theta_1$ , the gravity term is zero.

$$\frac{\partial L}{\partial \dot{\theta}_1} = [I_{zz1} + m_2 r_2^2 c_2^2 + m_3 (l_2 c_2 + r_3 c_{23})^2] \dot{\theta}_1 \quad (37)$$

Let effective inertia to be:

$$M_{11} = I_{zz1} + m_2 r_2^2 c_2^2 + m_3 (l_2 c_2 + r_3 c_{23})^2 \quad (38)$$

This will generate terms involving  $\dot{\theta}_2$  and  $\dot{\theta}_3$  (Coriolis/ Centrifugal effects).

$$\tau_1 = M_{11} \ddot{\theta}_1 + \dot{M}_{11} \dot{\theta}_1 \quad (39)$$

For joint 2  $\theta_2$  motion, collect all terms with  $\dot{\theta}_2$ :

$$\frac{\partial L}{\partial \dot{\theta}_2} = (I_2 + m_2 r_2^2 + m_3 l_2^2 + I_3 + m_3 r_3^2 + 2m_3 l_2 r_3 c_3) \dot{\theta}_2 + (I_3 + m_3 r_3^2 + m_3 l_2 r_3 c_3) \dot{\theta}_3 \quad (40)$$

The potential energy  $P$  and the  $c_2$  terms in  $K$  are differentiated.

$$\frac{\partial P}{\partial \theta_2} = g(m_2 r_2 c_2 + m_3 l_2 c_2 + m_3 r_3 c_{23}) \quad (41)$$

Hence, the gravity terms  $G_2 = \frac{\partial P}{\partial \theta_2}$ .

For joint 3  $\theta_3$  motion,

$$\frac{\partial L}{\partial \dot{\theta}_3} = (I_3 + m_3 r_3^2 + m_3 l_2 r_3 c_3) \dot{\theta}_2 + (I_3 + m_3 r_3^2) \dot{\theta}_3 \quad (42)$$

$$G_3 = \frac{\partial P}{\partial \theta_3} = m_3 g r_3 \cos(\theta_2 + \theta_3) \quad (43)$$

The equations are typically arranged in the form:

$$M(\theta) \ddot{\theta} + V(\theta, \dot{\theta}) + G(\theta) = \tau \quad (44)$$

Inertia matrix  $M(\theta)$  is given as:

$$M = \begin{bmatrix} M_{11} & 0 & 0 \\ 0 & M_{22} & M_{23} \\ 0 & M_{32} & M_{33} \end{bmatrix} \quad (45)$$

$$M_{11} = I_{zz1} + m_2 r_2^2 c_2^2 + m_3 (l_2 c_2 + r_3 c_{23})^2 \quad (46)$$

$$M_{22} = I_2 + m_2 r_2^2 + I_3 + m_3 (l_2^2 + r_3^2 + 2l_2 r_3 c_3) \quad (47)$$

$$M_{23} = M_{32} = I_3 + m_3 (r_3^2 + l_2 r_3 c_3) \quad (48)$$

$$M_{33} = I_3 + m_3 r_3^2 \quad (49)$$

Gravity Vector  $G(\theta)$  is given as:

$$G = \begin{bmatrix} 0 \\ g(m_2 r_2 c_2 + m_3 l_2 c_2 + m_3 r_3 c_{23}) \\ g(m_3 r_3 c_{23}) \end{bmatrix} \quad (50)$$

The Coriolis/ Centrifugal Vector  $V(\theta, \dot{\theta})$  is given as:

$$V = C(\theta, \dot{\theta}) \dot{\theta} \quad (51)$$

$$C(\theta, \dot{\theta}) = \begin{bmatrix} 0 & \frac{\partial M_{11}}{\partial \theta_2} \dot{\theta}_1 & \frac{\partial M_{11}}{\partial \theta_3} \dot{\theta}_1 \\ -\frac{1}{2} & -h \dot{\theta}_3 & -h(\dot{\theta}_2 + \dot{\theta}_3) \\ -\frac{1}{2} \frac{\partial M_{11}}{\partial \theta_3} \dot{\theta}_1 & g \dot{\theta}_2 & 0 \end{bmatrix} \quad (52)$$

Where:

$$h = m_3 l_2 r_3 \sin(\theta_3) \quad (53)$$

$$\frac{\partial M_{11}}{\partial \theta_2} = -[m_2 r_2^2 \sin(2\theta_2) + m_3 (l_2^2 \sin(2\theta_2) + r_3^2 \sin(2\theta_{23}) + 2l_2 r_3 \sin(2\theta_2 + \theta_3))] \quad (54)$$

$$\frac{\partial M_{11}}{\partial \theta_3} = -m_3 [r_3^2 \sin(2\theta_{23}) + 2l_2 r_3 \cos(\theta_2) \sin(\theta_{23})] \quad (55)$$

### 3.4 Design and Simulation of Robot Trajectory Control

The trajectory control scheme for the surgical robot manipulator integrates Resolved Acceleration Control (RAC) for kinematic precision and Active Force Control (AFC) for dynamic robustness. The hybrid strategy enables accurate end-effector motion by addressing both trajectory tracking and disturbance compensation. In this architecture, RAC forms the outer (kinematic) control loop. The output of the loop gives reference accelerations for the manipulator based on position and velocity errors. AFC acts as the

inner (dynamic) control loop, which acts to reject disturbances and unmodelled dynamics by compensating applied torques in real-time.

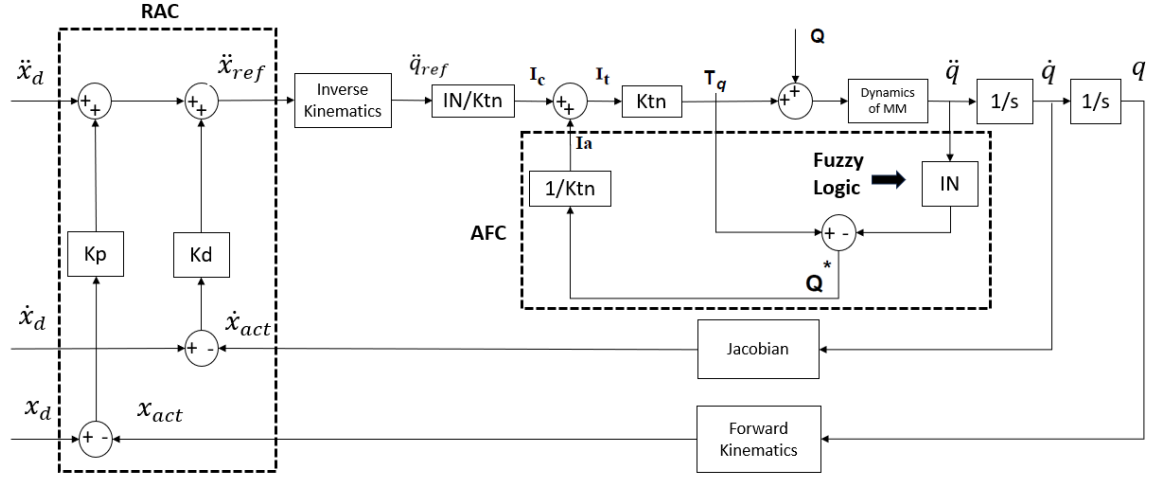


Figure 3.7: Fuzzy AFC Control Schematic with RAC

### 3.4.1 RAC Control Strategy

The Resolve Acceleration Control (RAC) forms the outer loop of the control scheme. RAC is a kinematic control loop designed to minimize position and orientation errors. Originally introduced, Resolved Acceleration Control (RAC) is a robust control technique that extends the principles of Resolved Motion Rate Control (RMRC) to the acceleration level (Shin & Lee, 1985). Unlike joint-space controllers (e.g., independent joint PID), which minimize error in joint angles ( $q$ ), RAC operates directly in the Cartesian task space ( $x, y, z$ ). The fundamental objective of RAC is to linearize the non-linear robotic system and decouple the end-effector motions. It computes the required joint accelerations ( $\ddot{q}_{ref}$ ) to ensure the end-effector tracks a reference trajectory ( $x_d$ ) with specified second-order error dynamics.

To ensure asymptotic tracking of the desired trajectory  $x_d(t)$ , we define an auxiliary control signal (reference Cartesian acceleration), denoted as  $\alpha$ , based on a Proportional-Derivative (PD) feedback law:

$$\ddot{x}_{ref} = \ddot{x}_d + K_d(\dot{x}_d - \dot{x}) + K_p(x_d - x) \quad (56)$$

Where:

- $\ddot{x}_d$  is the feedforward desired acceleration.
- $e = x_d - x$  is the position error.
- $\dot{e} = \dot{x}_d - \dot{x}$  is the velocity error.
- $K_p$  and  $K_d$  are positive definite gain matrices (typically diagonal).

Substituting  $\ddot{x} = \alpha$  into the error equation yields the closed-loop error dynamics:

$$\ddot{e} = K_d\dot{e} + K_p e = 0 \quad (57)$$

This represents a stable, second-order linear differential equation, ensuring that the error converges to zero as  $t \rightarrow \infty$ .

The output of the RAC serves as the input command to the AFC inner loop, which ensures dynamic compensation for external loads, such as tissue interaction forces and actuator disturbances.

### 3.4.2 AFC Control Strategy

Active force control (AFC) mechanism forms the inner loop of the proposed hybrid RAC-AFC structure and operates at the dynamic level of the robot manipulator. AFC compensates for external disturbances, model uncertainties, and nonlinearities inherent in the manipulator dynamics.

AFC is based on Newton's second law of motion, which relates the total torque applied to a rotating body to its angular acceleration and inertia. For a robotic joint, this can be expressed as:

$$\hat{\tau}_d = \tau_m - IN\ddot{q}_m \quad (58)$$

Where:

- $\tau_m$  is the measured actuator torque (obtained via torque or current sensing),
- $\ddot{q}_m$  is the measured joint acceleration, and
- $IN$  is the estimated inertia matrix or its equivalent scalar value for each joint.

The estimated disturbance  $\hat{\tau}_d$  is then positively fed back into the actuator command to produce the corrected torque:

$$\tau_c = \tau_{\text{ref}} + \hat{\tau}_d \quad (59)$$

This feedback compensates for unwanted external effects and maintains dynamic equilibrium, ensuring that the joint follows the acceleration commands derived from the outer RAC loop.

Accurate estimation of the inertia parameter  $IN$  is vital for effective disturbance rejection. In this study,  $IN$  is estimated using the adaptive scheme, Fuzzy Logic to accommodate changes in system configuration and loading.

The AFC loop directly interfaces with the RAC-generated reference accelerations. Substituting the reference acceleration  $\ddot{q}_{\text{ref}}$  from the RAC law into the AFC torque equation yields:

$$\tau_c = IN[\ddot{q}_{\text{ref}} - \ddot{q}_{\text{act}}] + \tau_m \quad (60)$$

where  $\ddot{q}_{\text{act}}$  is the measured joint acceleration. Equation (60) effectively integrates the kinematic command (RAC) with dynamic compensation (AFC), which producing a robust torque control signal capable of suppressing both internal model errors and external perturbations.

In hardware implementation, AFC relies on the torque/current sensors on joint actuators to measure  $\tau_m$ , MEMS accelerometers to capture  $\ddot{\theta}_m$ , and a microcontroller or

FPGA executing Equations (56) – (60) at high frequency ( $\geq 1$  kHz) for real-time compensation. The control parameters  $IN$ ,  $K_p$ , and  $K_d$  are fine-tuned experimentally to achieve high stiffness without overshoot, ensuring steady, compliant, and accurate surgical manipulation.

### 3.4.3 Inertia Matrix Estimation using Fuzzy Logic

To address the limitations of the standard Active Force Control (AFC) scheme, this project implements an adaptive tuning mechanism using Fuzzy Logic. In the classical AFC formulation, the estimated inertia matrix ( $IN$ ) is a fixed parameter. However, for a 3-DOF RRR manipulator, the effective inertia varies significantly with configuration (e.g., changes in the moment of inertia as the arm extends) and in the presence of unmodeled disturbances. A fixed  $IN$  often results in a trade-off: a high value improves disturbance rejection but introduces "chattering" during steady-state tracking, while a low value ensures smooth motion but sluggish reaction to external forces.

To resolve this, a Fuzzy Inference System (FIS) was designed to adjust the diagonal elements of the  $IN$  matrix in real-time. The objective is to keep the controller compliant (low gain) when error is minimal to save energy and reduce vibration, but stiff (high gain) when tracking errors occur.

A Mamdani-type fuzzy controller was developed with two inputs and one output. The system evaluates the state of each joint independently. The inputs of FIS are  $e$  and  $\dot{e}$ , with the membership function of negative, zero and positive. The output is set as Estimated Inertia ( $IN$ ), which is defined over the range  $[0.1, 1.0]$ . A lower limit of 0.1 was enforced to ensure the AFC loop remains active even during low-error states. Three Triangular MFs were used:

1. Small (Centered at 0.1): For steady-state tracking.
2. Medium (Centered at 0.5): For nominal operation.
3. Large (Centered at 1.0): For aggressive disturbance rejection.

The fuzzy rules were constructed to map the error dynamics to the required control stiffness. The logic follows the physical principle that a larger inertia estimate in the AFC loop results in a larger counter-torque calculation. The rules are as follows:

1. If  $e$  is 0 and  $e_{dot}$  is 0, then  $IN$  is small. (This minimizes noise amplification when the robot is on the correct trajectory.)
2. If  $e$  is large (+ve/-ve), then  $IN$  is large. (This triggers an immediate increase in corrective torque to reject the disturbance.)
3. If  $e_{dot}$  is large, then  $IN$  is large (This anticipates future position deviation and stiffens the joint before the error accumulates.)

The FIS is embedded within the inner AFC loop and evaluated at every sampling interval. The adaptive inertia matrix is updated dynamically as:

$$IN(t) = \text{diag}[FIS(e_1, \dot{e}_1), FIS(e_2, \dot{e}_2), FIS(e_3, \dot{e}_3)] \quad (61)$$

This adaptive value is then fed into the torque estimation equation:

$$T_{dist}(t) = T_{meas}(t) - IN(t) \cdot \ddot{q}_{est}(t) \quad (62)$$

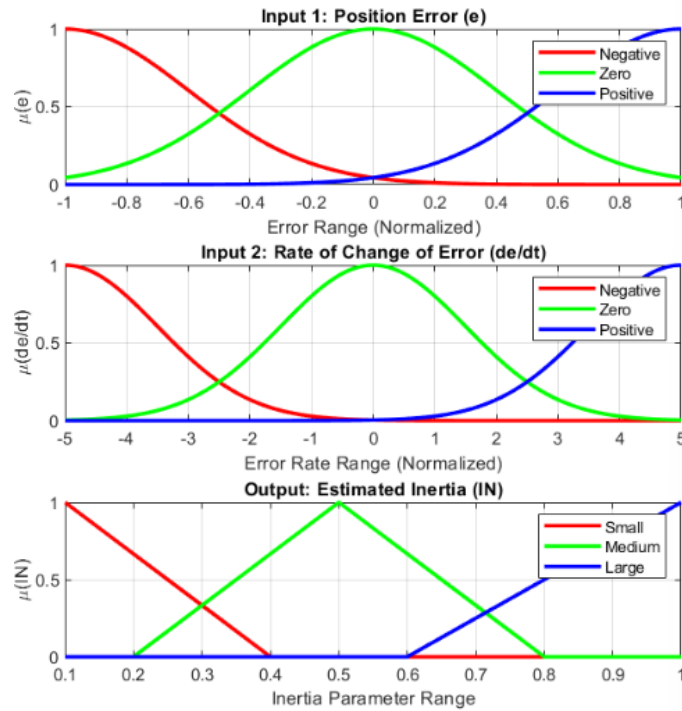


Figure 3.8: Fuzzy Inference System

### **3.5 Testing Robotic System with Surgical Operation**

#### **3.5.1 Simulation**

A performance baseline for the proposed control architecture prior to physical deployment, comprehensive verification was conducted within a numerical simulation environment using MATLAB and Simulink. The simulation focuses exclusively on evaluating the end-effector's translational motion along the X, Y, and Z axes, assessing the kinematic tracking loop governed by the Resolved Acceleration Control (RAC) alongside the dynamic disturbance rejection of the Active Force Control (AFC) loop.

To simulate a realistic clinical task, the 3-DOF RRR manipulator model was subjected to an automated single-plane zigzag suturing trajectory. This reference pathway mathematically couples a constant linear scanning progression along the incision line (X-axis) with a continuous, high-frequency sinusoidal stitching wave (Y-axis) at a fixed height profile. To replicate the unstructured conditions of an operational theatre, a time-varying external disturbance torque was mathematically injected directly into the plant dynamics. This disturbance acts as a proxy for unmodeled joint friction, tissue interactions, and high-frequency surgeon hand tremors, ensuring a rigorous evaluation of the Fuzzy-AFC loop's real-time compensation capabilities.

#### **3.5.2 Reference trajectory: Zigzag Suturing Pathway**

To emulate a realistic surgical operation, the manipulator will be commanded to follow a single-plane zigzag trajectory. This pathway was selected because it mimics the repetitive motion of a suturing task, which requires precise directional changes and sharp cornering. To specify, the trajectory path consists of a series of diagonal linear segments constrained to a 2D plane within the robot's 3D workspace. The end-effector must maintain a constant velocity along the linear segments and decelerate/accelerate appropriately at the turning points to minimize inertial vibration.

During each run, the real-time position of the end-effector will be logged via the AS5048A absolute encoders. The instantaneous joint angles ( $\theta$ ) will be recorded at a sampling rate of 100 Hz (or higher) and converted to Cartesian coordinates ( $x, y, z$ ) using the Forward Kinematics model, as the measured coordinates ( $x_{actual}, y_{actual}, z_{actual}$ ).

### 3.6 Validation and Comparison with Other Works

#### 3.6.1 Tracking Error

To evaluate the precision of the Intelligent Active Force Control (AFC) scheme, the tracking error is determined for both the simulation and the experimental runs. The position tracking error vector,  $e(t)$ , is defined as the deviation between the desired reference trajectory and the actual measured position of the end-effector at any given time instant  $t$ :

$$e_x(t) = x_{ref}(t) - x_{actual}(t) \quad (63)$$

$$e_y(t) = y_{ref}(t) - y_{actual}(t) \quad (64)$$

$$e_z(t) = z_{ref}(t) - z_{actual}(t) \quad (65)$$

The collected data will be plotted to visualize the trajectory deviation over time.

#### 3.6.2 Comparative Assessment

To validate the superiority of the proposed scheme, a comparative study is conducted. The performance of the Intelligent AFC controller is benchmarked against three baseline control strategies. First is the comparison with the standard RAC only controller, in order to demonstrate the baseline performance without force compensation. Second is the comparison with the conventional AFC only controller, so that to highlight the improvements gained by adding the “Intelligent” tuning mechanism.

## CHAPTER 4: RESULTS & DISCUSSION

### 4.1 Introduction

The primary objective of this section is to demonstrate the logical progression of the research, transitioning from the theoretical design and physical realization of the robotic system to the rigorous verification of its control capabilities. The discussion begins by establishing the physical constraints and capabilities of the hardware through an analysis of the robotic manipulator's design specifications, including the selection of actuators based on payload requirements and the results of the joint torque calculations. This provides the necessary context for the subsequent mathematical modelling results, where the derivation of the Denavit-Hartenberg (D-H) parameters and the formulation of the kinematic and dynamic equations are presented to define the equations of motion that govern the system's behavior.

Following the establishment of the physical and mathematical models, the chapter focuses on the core contribution of this research: the design and simulation of the trajectory control strategies within the MATLAB/Simulink environment. This section details the step-by-step implementation of the Resolved Acceleration Control (RAC) as the baseline motion controller, followed by the integration of the Active Force Control (AFC) loop designed to compensate for external disturbances inherent in surgical procedures. The chapter concludes with a critical validation and comparative analysis of the system's performance, where the tracking error and disturbance rejection capabilities of the proposed Intelligent AFC scheme are quantitatively benchmarked against standard RAC, conventional AFC and PID methods. This comparative evaluation highlights the specific improvements achieved by the proposed intelligent scheme, providing empirical evidence of its superior accuracy and robustness in maintaining precise end-effector translation.

## **4.2 Result of Design of Robotic Arm**

### **4.2.1 Design Model of Robot Manipulator**

The structural framework of the surgical manipulator was first conceptualized and modeled as a multi-variable serial-chain system using SolidWorks computer-aided design (CAD) software, as designated in Figure 4.1. To derive the precise coordinate transformations required for task-space operations, Denavit-Hartenberg (DH) frame assignments were established globally across the joint configurations.

The physical parameters and link lengths (such as the distance of  $L_3$  verified via the CAD environment in Figure 4.3) define the reachable workspace envelope of the end-effector. The exact measured values used to construct the kinematic matrices are summarized in Table 4.1.

To implement the design physically, the rigid links of the manipulator were fabricated using Fused Deposition Modeling (FDM) 3D printing with Polylactic Acid (PLA) filament. PLA was selected because of its highly advantageous strength-to-weight ratio. Producing a lightweight physical system effectively minimizes the overall inertial mass and reduces the gravitational load applied to the joints. This material choice directly eases the dynamic compensation burden on the inner Active Force Control (AFC) loop during rapid trajectory adjustments.

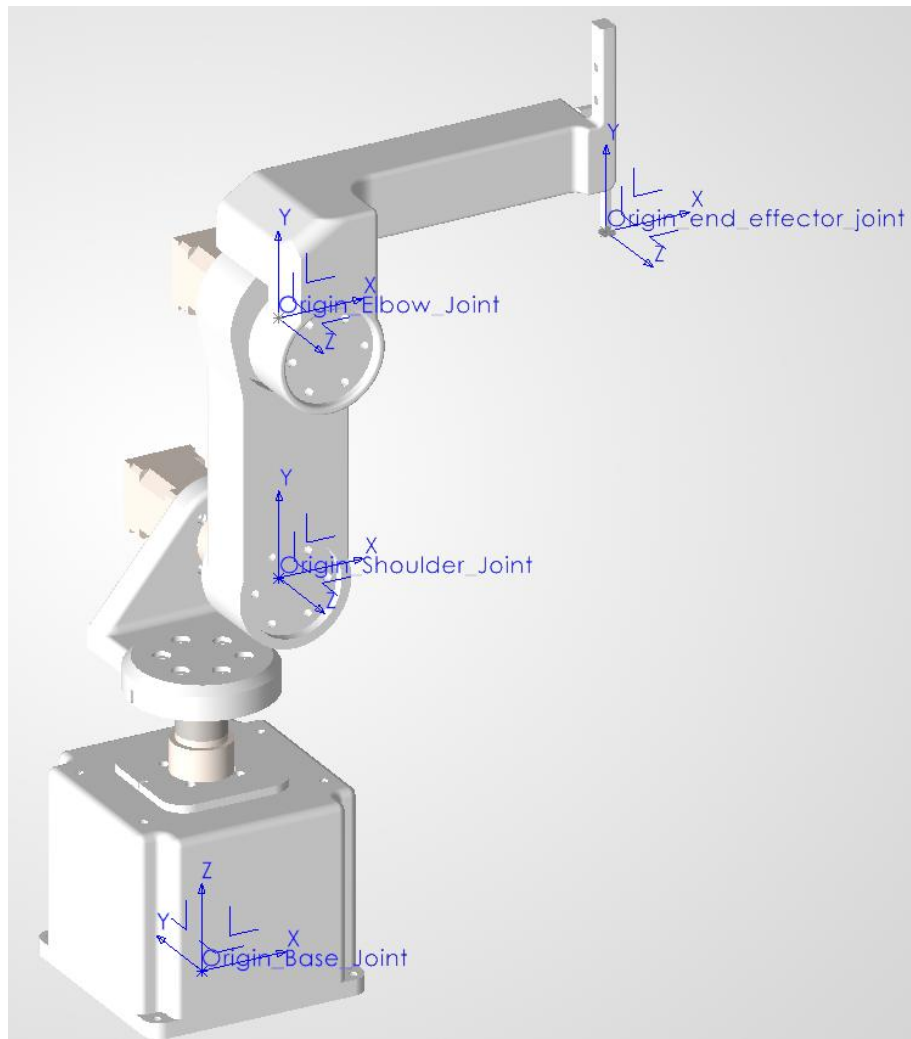


Figure 4.1: Frame Assignment of 3-DoF Robotic Arm Design

Table 4.1: D-H Parameters Measured

| $i$ | $a_i$             | $\alpha_i$         | $d_i$            | $\theta_i$ |
|-----|-------------------|--------------------|------------------|------------|
| 1   | 49.23 mm          | $90^\circ (\pi/2)$ | $L_1 = 231.9$ mm | $q_1$      |
| 2   | $L_2 = 160.00$ mm | $0^\circ$          | 0                | $q_2$      |
| 3   | $L_3 = 210.87$ mm | $0^\circ$          | $L_4 = 3$ mm     | $q_3$      |



Figure 4.2: Fabricated Robotic Arm for Testing

#### 4.2.2 Torque Calculation and Actuator Selection

To ensure the manipulator is capable of maintaining precise trajectory tracking without saturation or stalling, a static torque analysis was conducted. The critical design condition was identified as the fully extended horizontal configuration. In this pose, the gravitational moment arms for the links are maximized, requiring the highest holding torque from the joint actuators.

The required torque was calculated based on the mass and length parameters of the fabricated links. The static torque for joints is governed by the mass of link(s) acting at its (their) centroid  $L/2$ ). Below are the calculations for static torque at respective joints:

$$\tau_{shoulder} = \tau_{Link\ 2} + \tau_{Link\ 3} \quad (66)$$

$$\tau_{shoulder} = m_2 g \left( \frac{L_2}{2} \right) + m_3 g \left( L_2 + \frac{L_3}{2} \right) \quad (67)$$

$$\tau_{shoulder} = (2.0)(9.81) \left( \frac{0.16}{2} \right) + (2.4)(9.81) \left( 0.210 + \frac{0.21087}{2} \right) = 8.9962\ N \cdot m \quad (68)$$

$$\tau_{elbow} = m_3 g \left( \frac{L_3}{2} \right) \quad (69)$$

$$\tau_{elbow} = (2.4)(9.81) \left( \frac{0.21087}{2} \right) = 2.4823 \text{ N} \cdot \text{m} \quad (70)$$

While the static calculation defines the holding requirement, during implementation we incorporate a Safety Factor (SF) to account for rotational inertia - torque required to accelerate the links from a standstill, joint friction - non-linear friction in the 3D-printed joints and gearbox assemblies, as well as the rapid acceleration - the requirement for the Active Force Control (AFC) scheme to react quickly to disturbances. Consequently, the selected actuators must provide a maximum torque significantly higher than the static value of 8.9962 Nm to ensure a robust dynamic response.

For the vertical base axis, the gravitational torque is theoretically zero ( $\tau_g = 0$ ) as the gravity vector is parallel to the axis of rotation. The required torque is entirely dominated by the inertial dynamics of the manipulator:

$$\tau_1 = J_{eq}(\theta)\ddot{\theta}_1 + C(\theta, \dot{\theta}) + \tau_{friction} \quad (71)$$

Given the large rotational inertia when the arm is fully extended, a higher torque capacity is required for rapid acceleration and to overcome the large moment of inertia of the extended arm. Table 4.2 summarizes the final selection of actuator models and their specifications.

The joint motor capacities were chosen following a static holding torque analysis optimized for the fully extended horizontal arm position, factored alongside a safe operational margin to accommodate non-linear joint friction and acceleration torques.

Table 4.2: Actuator Specifications and Torque Output

| Joint<br>Axis | Stepper<br>Motor<br>Model | Rated<br>Torque | Gear<br>Reduction | Effective<br>Output<br>Torque | Holding Dominance |
|---------------|---------------------------|-----------------|-------------------|-------------------------------|-------------------|
|               |                           |                 |                   |                               |                   |

|                       |             |         |      |          |                                                                     |
|-----------------------|-------------|---------|------|----------|---------------------------------------------------------------------|
| Base<br>(Joint 1)     | NEMA-<br>17 | 2.30 Nm | 20:1 | 46.00 Nm | Overcomes high rotational inertia of fully extended configurations. |
| Shoulder<br>(Joint 2) | NEMA-<br>17 | 0.52 Nm | 19:1 | 9.88 Nm  | Defends against peak gravitational moments.                         |
| Elbow<br>(Joint 3)    | NEMA-<br>17 | 0.52 Nm | 14:1 | 7.28 Nm  | Manages internal link mass and tool-tip constraints.                |

### 4.3 Results of Kinematics and Dynamics

#### 4.3.1 Forward Kinematics

The final calculated matrix with subjected model parameter values is substituted into the matrix shown below:

$$T_0^3 = \begin{bmatrix} c_1 c_{23} & -c_1 s_{23} & s_1 & c_1 (L_2 c_2 + L_3 c_{23}) \\ s_1 c_{23} & -s_1 s_{23} & -c_1 & s_1 (L_2 s_2 + L_3 c_{23}) \\ s_{23} & c_{23} & 0 & L_2 s_2 + L_3 s_{23} + L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (72)$$

Where the Position Vector (P) are given as:

$$P_x = \cos(q_1) [210.87 \cos(q_2 + q_3) + 160 \cos(q_2) + 49.23] \quad (73)$$

$$P_y = \sin(q_1) [210.87 \cos(q_2 + q_3) + 160 \cos(q_2) + 49.23] \quad (74)$$

$$P_z = 210.87 \sin(q_2 + q_3) + 160 \sin(q_2) + 231.9 \quad (75)$$

Table 4.3: D-H Parameters Measured

| $i$ | $a_i$             | $\alpha_i$         | $d_i$            | $\theta_i$ |
|-----|-------------------|--------------------|------------------|------------|
| 1   | 49.23 mm          | $90^\circ (\pi/2)$ | $L_1 = 231.9$ mm | $q_1$      |
| 2   | $L_2 = 160.00$ mm | $0^\circ$          | 0                | $q_2$      |

|   |                         |           |                     |       |
|---|-------------------------|-----------|---------------------|-------|
| 3 | $L_3 = 210.87\text{mm}$ | $0^\circ$ | $L_4 = 3\text{ mm}$ | $q_3$ |
|---|-------------------------|-----------|---------------------|-------|

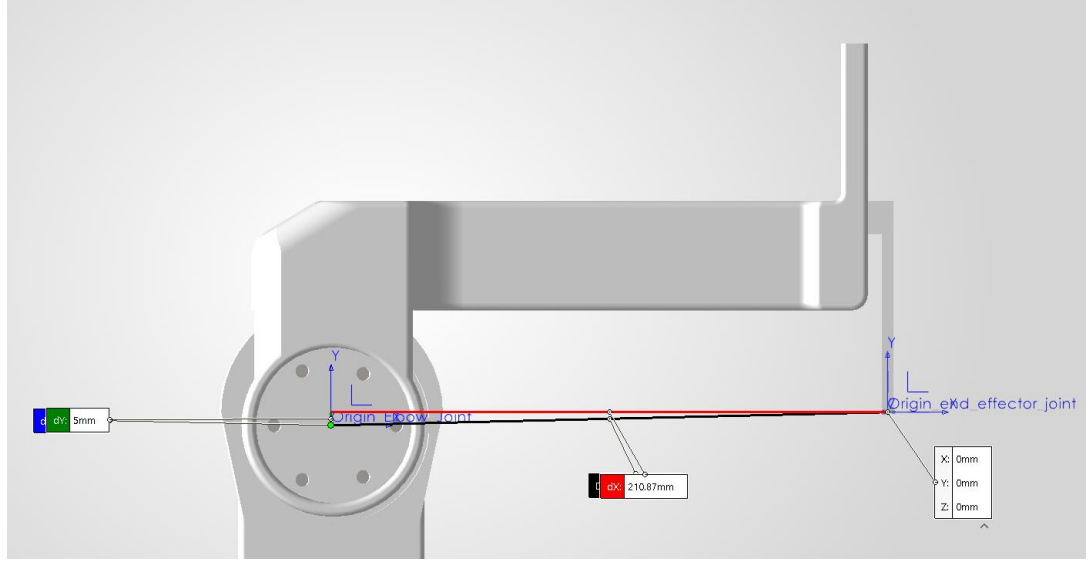


Figure 4.3: Distance of  $L_3$  Measured in SolidWorks Model

(Note: In real hardware implementation:  $L_3$  represents normal distance between elbow-end-effector coordinates.)

#### 4.3.2 Dynamics

Based on the model parameters, we substitute the values into the 3-DOF RRR dynamics equations. We treat the links as uniform slender rods and the base as a solid cylinder. COM is located at the midpoint of each link ( $r = \frac{L}{2}$ ). Inertia is modeled as  $I = \frac{1}{12}mL^2$  for links and  $I = mr^2$  for the base.

Input values are as follows:

1. Link 2:  $L_2 = 0.16\text{ m}$ ,  $m_2 = 2.0\text{ kg}$ ,  $r_2 = \frac{0.16}{2} = 0.08\text{ m}$

$$I_2 = \frac{1}{12}(2.0)(0.16)^2 = 0.004267\text{ kg} \cdot \text{m}^2 \quad (76)$$

2. Link 3:  $L_3 = 0.21087\text{ m}$ ,  $m_3 = 2.4\text{ kg}$ ,  $r_3 = \frac{0.21087}{2} = 0.105435\text{ m}$

$$I_3 = \frac{1}{12}(2.4)(0.21087)^2 = 0.008893\text{ kg} \cdot \text{m}^2 \quad (77)$$

3. Base Cylinder:  $m_{base} = 1.1 \text{ kg}$ ,  $r_{base} = 0.045 \text{ m}$

$$I_{zz1} = \frac{1}{2}(1.1)(0.045)^2 = 0.001114 \text{ kg} \cdot \text{m}^2 \quad (78)$$

Substitute into the Inertia Matrix  $M(\theta)$  of Equation (45), we obtained:

Waist Inertia:

$$M_{11} = I_{zz1} + m_2 r_2^2 \cos^2 \theta_2 + m_3 (L_2 \cos \theta_2 + r_3 \cos(\theta_2 + \theta_3))^2 \quad (79)$$

$$M_{11} = 0.001114 + 2.0(0.08)^2 \cos^2 \theta_2 + 2.4(0.16 \cos \theta_2 + 0.105435 \cos(\theta_2 + \theta_3))^2 \quad (80)$$

$$M_{11} = 0.001114 + 0.0128 \cos^2 \theta_2 + 2.4 (0.16 \cos \theta_2 + 0.105435 \cos(\theta_2 + \theta_3))^2 \quad (81)$$

Shoulder Inertia:

$$M_{22} = I_2 + I_3 + m_2 r_2^2 + m_3 L_2^2 + m_3 r_3^2 + 2m_3 L_2 r_3 \cos \theta_3 \quad (82)$$

$$M_{22} = 0.004267 + 0.008893 + 2.0(0.08)^2 + 2.4(0.16)^2 + 2.4(0.105435)^2 + 2(2.4)(0.16)(0.105435) \cos \theta_3 \quad (83)$$

$$M_{22} = 0.11408 + 0.080974 \cos \theta_3 \quad (84)$$

Coupling Inertia:

$$M_{23} = M_{32} = I_3 + m_3 r_3^2 + m_3 L_2 r_3 \cos \theta_3 \quad (85)$$

$$M_{23} = 0.008893 + 2.4(0.105435)^2 + (2.4)(0.16)(0.105435) \cos \theta_3 \quad (86)$$

$$M_{23} = 0.035573 + 0.040487 \cos \theta_3 \quad (87)$$

Elbow Inertia:

$$M_{33} = I_{3_{joint}} + m_3 r_3^2 = 0.008893 + 0.02668 = 0.035573 \quad (88)$$

Substituting the parameter values into the Gravity Vector  $G(\theta)$  in Equation (50), we obtained:

$$G(\theta) = \begin{bmatrix} 0 \\ 5.3366 \cos(\theta_2) + 2.4824 \cos(\theta_2 + \theta_3) \\ 2.4824 \cos(\theta_2 + \theta_3) \end{bmatrix} \quad (89)$$

We define an auxiliary variable representing part of the  $M_{11}$  scaling architecture:

$$K_{term} = 2.4(0.16 \cos \theta_2 + 0.105435 \cos \theta_{23}) \quad (90)$$

The non-zero partial derivatives of  $M_{11}$  with respect to joint geometry coordinate changes are:

$$\frac{\partial M_{11}}{\partial \theta_2} = -0.0128 \sin(2\theta_2) - 2K_{term}(0.16 \sin \theta_2 + 0.105435 \sin \theta_{23}) \quad (91)$$

$$\frac{\partial M_{11}}{\partial \theta_3} = -2K_{term}(0.105435 \sin \theta_{23}) \quad (92)$$

$$C_{12} = \frac{\partial M_{11}}{\partial \theta_2} \dot{\theta}_1 \quad (93)$$

$$C_{13} = \frac{\partial M_{11}}{\partial \theta_3} \dot{\theta}_1 \quad (94)$$

$$C_{21} = -\frac{1}{2} C_{12} \quad (95)$$

$$C_{31} = -\frac{1}{2} C_{13} \quad (96)$$

For the remaining link coupling parameters, the common term  $h$  is defined as:

$$h = m_3 L_2 r_3 \sin \theta_3 = 2.4(0.16)(0.105435) \sin \theta_3 = 0.040487 \sin \theta_3 \quad (97)$$

$$C_{22} = -h \dot{\theta}_3 = -0.040487 \sin(\theta_3) (\dot{\theta}_3) \quad (98)$$

$$C_{23} = -h(\dot{\theta}_2 + \dot{\theta}_3) = -0.040487 \sin(\theta_3) (\dot{\theta}_2 + \dot{\theta}_3) \quad (99)$$

$$C_{32} = h \dot{\theta}_2 = 0.040487 \sin(\theta_3) (\dot{\theta}_2) \quad (100)$$

The final dynamics for specific parameters are:

Joint 1 (Waist):

$$M_{11}\ddot{\theta}_1 + \left(\frac{\partial M_{11}}{\partial \theta_2}\dot{\theta}_2 + \frac{\partial M_{11}}{\partial \theta_3}\dot{\theta}_3\right)\dot{\theta}_1 = \tau_1 \quad (101)$$

Joint 2 (Shoulder):

$$(0.11408 + 0.080974c_3)\ddot{\theta}_2 + (0.035573 + 0.040487c_3)\ddot{\theta}_3 - \frac{1}{2}\frac{\partial M_{11}}{\partial \theta_2}\dot{\theta}_1^2 - 0.040487s_3\dot{\theta}_3^2 - 0.080874s_3\dot{\theta}_2\dot{\theta}_3 + G_2 = \tau_2 \quad (102)$$

Joint 3 (Elbow):

$$(0.035573 + 0.040487c_3)\ddot{\theta}_2 + 0.035573\ddot{\theta}_3 - \frac{1}{2}\frac{\partial M_{11}}{\partial \theta_3}\dot{\theta}_1^2 + 0.040487s_3\dot{\theta}_2^2 + G_3 = \tau_3 \quad (103)$$

## 4.4 Results of Design and Simulation of Trajectory Control

### 4.4.1 Setup of Control Simulation

The proposed Fuzzy AFC with RAC control scheme was implemented and verified using a numerical simulation environment developed in MATLAB. The simulation was designed to test the controller's robustness in trajectory tracking tasks relevant to surgical applications, specifically a suturing motion.

The simulation operates on a fixed-step discrete time basis to emulate the behavior of a digital embedded controller (e.g., Raspberry Pi). Solver type of Fixed-Step Discrete is used with a sampling time ( $dt$ ) of 0.001 seconds (1 kHz). Simulation Duration (T) is set for 4.0 seconds. The plant model represents a 3-DOF RRR (Revolute-Revolute-Revolute) manipulator. The Denavit-Hartenberg (DH) parameters and dynamic properties (mass) used in the simulation are as calculated in Section 4.2 and 4.3.

To simulate a surgical suturing procedure, a “Zig-Zag” trajectory was mathematically generated. The path combines a linear scanning motion along the incision (X-axis) with a sinusoidal stitching motion (Y-axis). The reference trajectory is defined as:

$$x_d(t) = x_{start} + v_{scan} \cdot t \quad (104)$$

$$y_d(t) = y_{start} + A_{stitch} \cdot \sin(\omega_{stitch} \cdot t) \quad (105)$$

$$z_d(t) = z_{height} \quad (106)$$

Where:

- Scan speed ( $v_{scan}$ ): 0.01 m/s
- Stitch amplitude ( $A_{stitch}$ ): 0.05 m (width)
- Stitch frequency ( $\omega_{stitch}$ ): 4.0 rad/s
- Reference height ( $z_{height}$ ): 0.2032 m

The controller architecture consists of a Resolved Acceleration Control (RAC) outer loop for kinematic error correction and an Active Force Control (AFC) inner loop, built upon a computed-torque (inverse-dynamics) law that feeds forward the estimated inertia, Coriolis, and gravity terms of the manipulator. Rather than employing an iterative inverse-kinematics solver, the manipulator starts directly from a fixed, repeatable home configuration. The home joint angles are set to  $q = [0, 0.6, -1.2]$  rad (base, shoulder, elbow), corresponding to a shoulder-raised, elbow-bent posture. This configuration places the end-effector at  $[0.3553, 0.0000, 0.2032]$  m and yields a well-conditioned Jacobian (condition number  $\approx 4.18$ ), avoiding the kinematic singularity associated with the fully-extended zero pose. The suturing trajectory is anchored at this start position so that the commanded path remains entirely within the reachable workspace.

A Mamdani-type Fuzzy Inference System (FIS) was configured to adaptively tune the effective inertia scale factor (IN) of the AFC loop in real time. The control gains used in the simulation are shown in Table 4.4.

Table 4.4: Control gains

| Controller | Parameters                      | Set Values                 |
|------------|---------------------------------|----------------------------|
| RAC        | Proportional Gain ( $K_p$ )     | $400 \cdot I_{3 \times 3}$ |
|            | Derivative Gain ( $K_d$ )       | $120 \cdot I_{3 \times 3}$ |
| AFC        | Torque Sensitivity ( $K_{tn}$ ) | 1.0                        |

To test the disturbance rejection capability of the AFC loop, a time-varying external disturbance torque  $T_{\text{dist}}$  was injected directly into the plant dynamics:

$$T_{\text{dist}} = [0.2 \sin(5t), 0.2 \cos(5t), 0.1]^T \text{ N} \cdot \text{m} \quad (107)$$

The simulation logs the Lyapunov function candidate  $V(z) = z^T P_z$  at every time step to verify the stability of the closed-loop system under these disturbance conditions.

#### 4.4.2 Fuzzy Rule Base

The fuzzy inference system utilizes two inputs: Position Error ( $e$ ) and Change of Error ( $\dot{e}$ ), and one output: the Estimated Inertia Parameter ( $IN$ ), ranging from 0.8 to 1.2. The output is a dimensionless multiplier applied to the model inertia term of the computed-torque law, rather than an absolute inertia value. The linguistic variables for the inputs and output are defined below.

The variables for inputs ( $e$  and  $\dot{e}$ ) are as follows:

- N (Negative): Significant deviation in the negative direction.
- Z (Zero): The system state is within the acceptable tolerance band (stable).
- P (Positive): Significant deviation in the positive direction.

The variables for output ( $IN$ ) are as follows:

- Small (S): Reduced inertia scale ( $\approx 0.8$ ). Applied when the system is in steady state ( $e \approx 0$  and  $\dot{e} \approx 0$ ) to keep the response smooth and prevent chatter during fine tracking.
- Medium (M): Nominal inertia scale ( $\approx 1.0$ ). Represents the baseline model-based compensation.
- Large (L): Increased inertia scale ( $\approx 1.2$ ). Applied in dynamic states where error exists ( $e \neq 0$  or  $\dot{e} \neq 0$ ) to boost robustness and disturbance rejection.

The rule base consists of 5 logic rules. When the tracking error and its rate are small the scale settles to its nominal value, whereas any growing position or velocity error raises the inertia scale to strengthen the compensating torque.

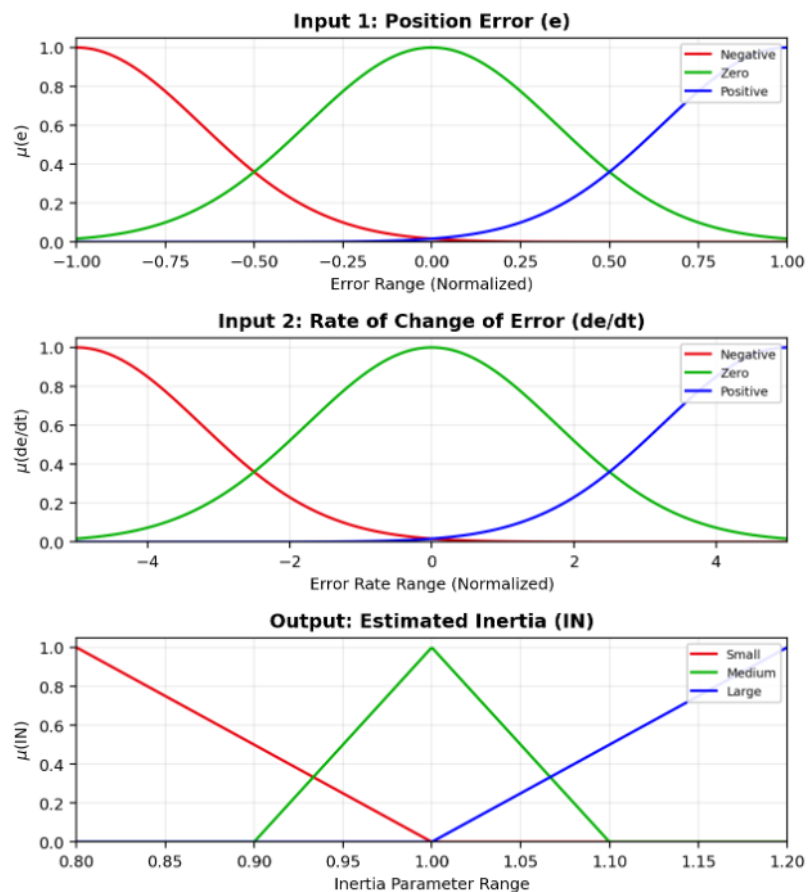


Figure 4.4: Membership Functions

Table 4.5: Logic rules of FIS

| Rule No. | IF Error (e) IS | AND Change of Error ( $\dot{e}$ ) IS | THEN Est. Inertia ( $IN$ ) IS | Rationale                                                 |
|----------|-----------------|--------------------------------------|-------------------------------|-----------------------------------------------------------|
| 1        | Zero            | Zero                                 | Medium                        | Steady-state: hold nominal gain for smooth fine tracking. |
| 2        | Negative        | (Any)                                | Large                         | Deviation: raise gain to resist disturbance.              |
| 3        | Positive        | (Any)                                | Large                         | Deviation: raise gain to resist disturbance.              |
| 4        | (Any)           | Negative                             | Large                         | High velocity error: increase damping/stiffness.          |
| 5        | (Any)           | Positive                             | Large                         | High velocity error: increase damping/stiffness.          |

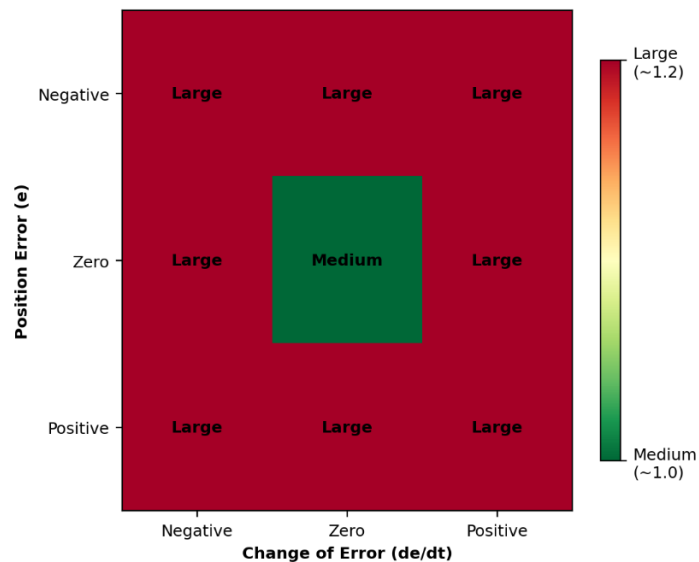


Figure 4.5: FIS Rule Base Mapping

## 4.5 Result of Testing and Experimental Work

### 4.5.1 System Components

Table 4.6: Summary of System Components

| System Subcomponent | Component Model/ Material     | Key Technical Specifications                          | Primary Functional Role in Experiment                             |
|---------------------|-------------------------------|-------------------------------------------------------|-------------------------------------------------------------------|
| Structural Links    | PLA Filament                  | FDM 3D Printed, Lightweight Profile                   | Forms physical RRR manipulator geometry.                          |
| Main Controller     | Raspberry Pi 3B+              | Broad Processing Power, Raspberry Pi OS lite (64-bit) | Executes RAC, Fuzzy FIS, and AFC loops simultaneously.            |
| Joint Actuators     | Stepper Motors                | High Torque Output                                    | Drives physical joint rotations based on commands.                |
| Position Sensors    | A5048A Magnetic Encoders      | 14-bit Resolution, 0.0219° Precision                  | Tracks joint angle (q) to close the outer RAC loop.               |
| Current Sensors     | Hall-Effect Sensors           | Analog/Digital In-line Feedback                       | Measures live current to isolate operational torque ( $\tau_m$ ). |
| Suturing Tooling    | Linear Reciprocating Actuator | Independent DC/Stepper Drive                          | Simulates vertical needle-driving and tissue puncture.            |

|                 |                     |                          |                                                                                      |
|-----------------|---------------------|--------------------------|--------------------------------------------------------------------------------------|
| System Software | Python<br>Scripting | 3.7<br><br>Sampling Rate | 100 Hz (or higher)<br><br>Manages real-time data logging and algorithmic processing. |
|-----------------|---------------------|--------------------------|--------------------------------------------------------------------------------------|

#### 4.5.2 System Initialization and Homing

Before commencing experimental trials, the robotic manipulator undergoes a standardized homing sequence to establish a reliable global reference frame. The home configuration is defined with the shoulder arm in a vertically extended position ( $90^\circ$  relative to the horizontal plane) and the elbow arm positioned parallel to the ground ( $0^\circ$ ). This posture ensures that all subsequent joint rotations are calculated from a repeatable, zero-displacement state.



Figure 4.6: Homing Starting Pose Configuration

#### 4.5.2 Kinematic Validation

The baseline coordinate transformation accuracy is verified through a two-stage process:

1. Forward Kinematics (FK): The system computes the theoretical end-effector position based on the measured encoder values.
2. Inverse Kinematics (IK): The system maps target Cartesian coordinates to the required joint angles.

During these tests, the position of the End-of-Arm Tooling (EOAT) is measured relative to the center of the base motor. This phase confirms that the mathematical model correctly translates control commands into physical spatial positioning, ensuring the arm is calibrated before the application of dynamic control laws.

The robot is tested to perform inverse kinematics, forward kinematics. The coordinate position of EAOT is measured from the center of the base motor.

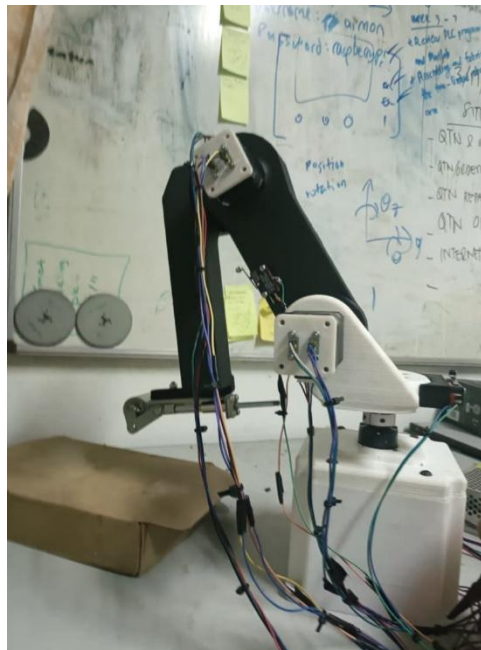


Figure 4.7: Inverse Kinematics Motion Control

#### 4.5.3 Control Algorithm Evaluation

To determine the efficacy of the proposed Intelligent Active Force Control (AFC) scheme, a comparative experimental study is conducted. The control system is subjected to three distinct testing configurations:

1. Resolved Acceleration Control (RAC) only: Establishes the baseline performance of the standard dynamic model.
2. Fuzzy-AFC only: Evaluates the capacity of the fuzzy logic loop to suppress disturbances without the compensatory benefits of the RAC framework.
3. Fuzzy-AFC-RAC: The proposed integrated controller, where the intelligence of the fuzzy tuner enhances the disturbance rejection of the RAC loop.

For each configuration, the system records the tracking performance for all three motors. The results are processed to calculate positional errors and are plotted as time-series graphs to visualize the stability and convergence of each control strategy.

#### **4.5.4 Trajectory Execution and Suturing Simulation**

The end-effector is programmed to execute a zig-zag trajectory, mirroring the path tested in the simulation environment. This trajectory mimics the essential spatial progression of a surgical suturing task.

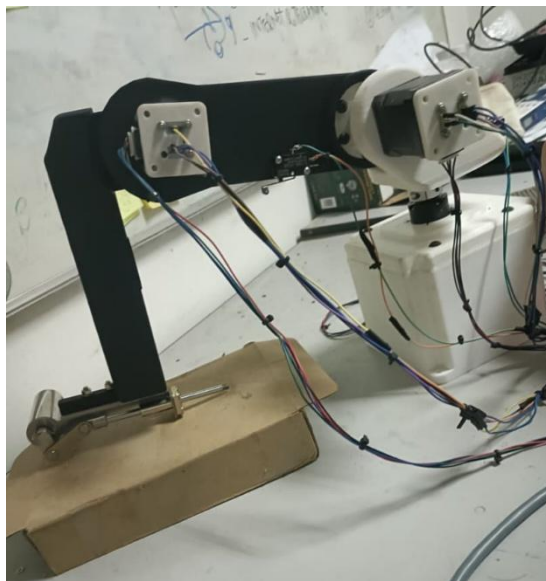


Figure 4.8: Path Starting Coordinate Position

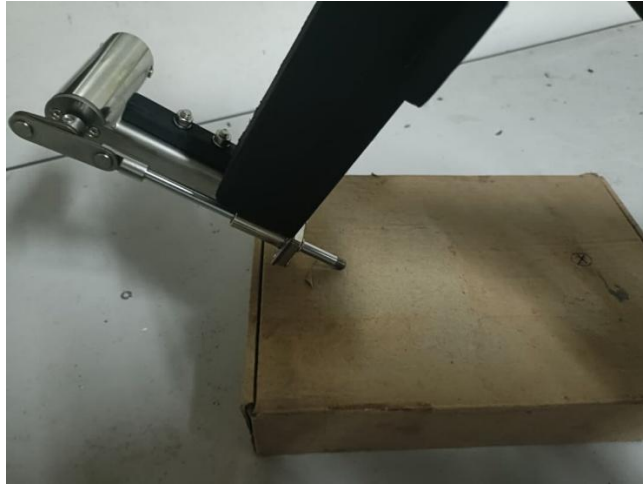


Figure 4.9: Path Ending Coordinate Position

To simulate the clinical suturing motion, the end-effector is equipped with a linear reciprocating actuator. This actuator drives a needle-mimicking element in a vertical, up-and-down motion synchronized with the arm's translational movement.

The end-effector is programmed to follow a zig-zag suturing trajectory path, similar to what's experiment in the simulation environment. The end-effector is equipped with a linear reciprocating actuator that mimic the needle up and down driving motion. The actuating code is attached in the appendix for further reference.



Figure 4.10: Zig-zag Path Starting Point 1



Figure 4.11: Zig-zag Path Peak Point 2



Figure 4.12: Zig-zag Path Peak Point 3



Figure 4.13: Zig-zag  
Path Peak Point 4



Figure 4.14: Linear Reciprocating Actuator  
for Needle Driving Motion

## 4.6 Result of Validation and Comparison with Existing Works

### 4.6.1 Simulation Analysis

The performance of the proposed Fuzzy AFC with RAC control scheme was evaluated using the zig-zag suturing trajectory, and the results are presented in Figure 4.9. The analysis focuses on position tracking accuracy, control effort, and the behaviour of the adaptive inertia parameter. Rather than relying on an iterative inverse-kinematics initialisation, the manipulator begins directly from a fixed, non-singular home configuration ( $q = [0, 0.6, -1.2]$  rad), which places the end-effector at  $[0.3553, 0.0000, 0.2032]$  m with a well-conditioned Jacobian (condition number  $\approx 4.18$ ). This avoids the singularity of the fully-extended pose and keeps the commanded path within the reachable workspace.

The position tracking error  $e$  (Figure 4.11) exhibits a stable, periodic behaviour corresponding to the frequency of the suturing motion (4.0 rad/s). The system maintains a bounded error throughout the trajectory despite the external disturbance, with a final recorded error of 1.096 mm (0.001096 m) and a Root Mean Square Error (RMSE) of 1.284 mm (0.001284 m). A brief initial transient peak of approximately 1.5 mm occurs

within the first 0.1 seconds as the robot accelerates from rest, after which the error settles into a low, bounded oscillation of roughly 1.0–1.5 mm. This residual ripple is expected, arising from the high-frequency nature of the zig-zag reference path combined with the injected sinusoidal disturbance, and it confirms the strong kinematic tracking capability of the RAC outer loop working together with the AFC compensation.

All control torques remained well within their safe operating bounds (Figure 4.12). The base joint (red) operates near  $\pm 2$  Nm, the shoulder joint (green) carries the largest sustained load at approximately 5–6 Nm due to gravity compensation, and the elbow joint (blue) remains close to 2 Nm. None of the joints approaches its limit ( $\pm 46$  Nm base,  $\pm 9.88$  Nm shoulder,  $\pm 7.28$  Nm elbow), and no torque saturation occurs at any point in the trajectory. The smooth, non-saturated torque profiles indicate that the computed-torque feed-forward (inertia, Coriolis, and gravity) supplies most of the required effort, leaving the feedback terms to handle only the residual error and disturbance. The trajectory is therefore mechanically feasible and safe for all three motors.

For the adaptive mechanism, the fuzzy-tuned inertia scale (IN) is active throughout the run. Starting from its nominal value, it settles to approximately 1.0007 and modulates slightly in response to the changing error state ( $e$ ,  $\dot{e}$ ). The small magnitude of this adaptation reflects that the model-based feed-forward already accounts for most of the dynamics, so the fuzzy loop is required only to fine-tune the inertia contribution rather than to perform large corrections. This confirms that the FIS is sensing the error state and adjusting the AFC response as intended.

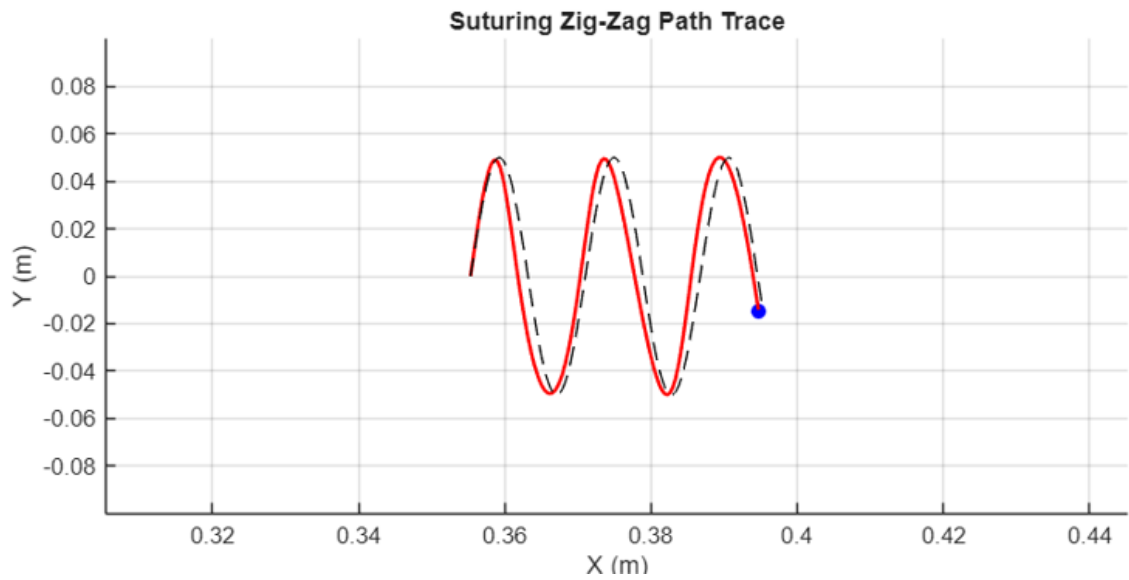


Figure 4.15: Path Trace of Fuzzy-AFC with RAC (Simulation)

```

Command Window
Home pose set. EE start position: [0.3553, -0.0000, 0.2032] m
Jacobian condition at home: 4.18 (well-conditioned)
Starting Simulation...

-----
SIMULATION RESULTS:
Final Error: 0.001096 m
Total RMSE: 0.001284 m
-----

```

Figure 4.16: Performance Metrics of Fuzzy-AFC with RAC (Simulation)

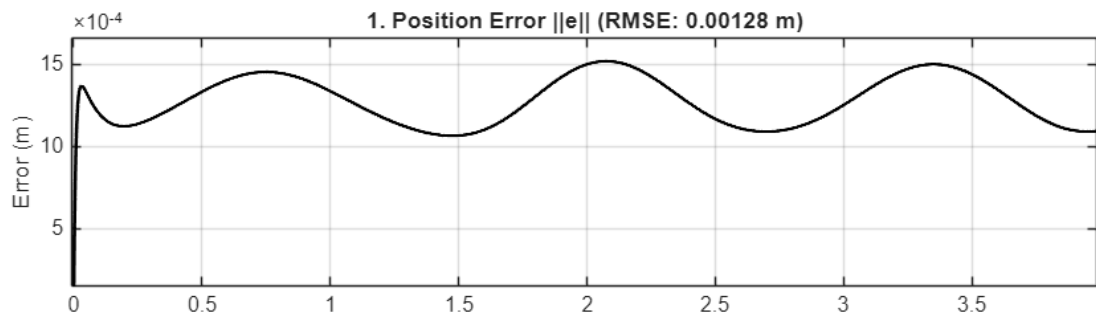


Figure 4.17: Position Error of Fuzzy-AFC with RAC (Simulation)

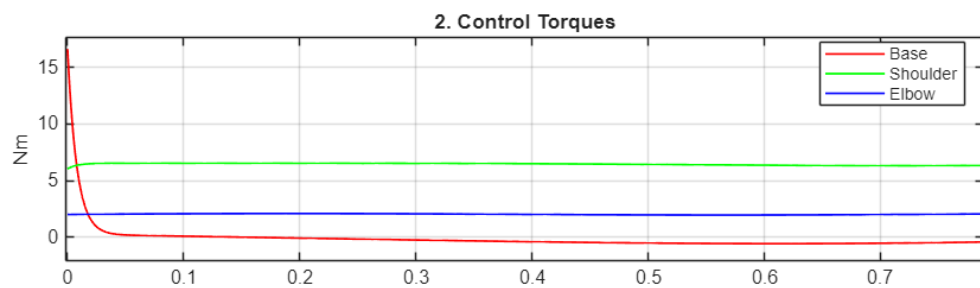


Figure 4.18: Control Torques of Fuzzy-AFC with RAC (Simulation)

Table 4.7: Control Gains' Effect on the Stability Result

| Parameter                          | Effect of Increasing                                                                                                                         | Effect of Decreasing                                                                                                            |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| $K_p$<br>(Proportional: Stiffness) | RMSE decreases Error (RMSE):<br>The robot tracks tighter. If too high (>200), it causes high-frequency vibrations and potential instability. | Error increases. Robot lags behind the zig-zag path.                                                                            |
| $K_d$ (Derivative: Damping)        | Smoother Motion. Oscillations and jitters are reduced. If too high, the system becomes slow to react and amplifies noise.                    | Oscillations increase.<br>Robot may overshoot corners in the zig-zag path.<br>The risk is system becomes underdamped and shaky. |

Table 4.8: Effect of  $K_{tn}$  on the Stability Result

| Scenario           | Effect on Result                                                                                                                                                                                                  |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ideal              | The estimated disturbance $Q_{est}$ is accurate. The AFC cancels forces perfectly.                                                                                                                                |
| Too high<br>(>1.0) | Over-Compensation. The controller thinks the motor is stronger than it is. It may command <i>less</i> current than needed, leading to larger position errors.                                                     |
| Too low<br>(<1.0)  | Aggressive Response. The controller thinks the motor is weak and pours in more current to compensate. This can improve tracking slightly but drastically increases Control Chattering, risking motor overheating. |

#### 4.6.1.1 Comparison with Baseline Controller

To quantify the contribution of each control component, the proposed integrated scheme was compared against the two baseline configurations defined in Section 4.5.3: a Resolved Acceleration Control (RAC) only loop, and a Fuzzy-AFC only loop driven by a joint-space PD reference. All three were tested on the identical trajectory, home pose, and disturbance for a fair comparison. The results are summarised in Table 4.9.

Table 4.9: Simulation Tracking Performance of the Three Control Configurations

| Control Configuration        | RMSE (mm) | Final Error (mm) | Peak Torque   |
|------------------------------|-----------|------------------|---------------|
| RAC only (baseline)          | 25.96     | 27.46            | Within limits |
| Fuzzy-AFC only<br>(baseline) | 5.91      | 4.25             | Within limits |
| Fuzzy-AFC-RAC<br>(proposed)  | 1.28      | 1.10             | Within limits |

The RAC-only configuration produces the largest error. Its tracking error climbs during the first second and then settles to a steady plateau of about 27 mm (RMSE 25.96 mm). Because this configuration applies only inertia weighting without active compensation of the Coriolis and gravity terms, the uncompensated joint loads — dominated by gravity sag at the shoulder — leave a persistent steady-state offset that the kinematic loop alone cannot remove. The path trace (Appendix B) consequently shows a visible vertical and phase offset from the reference.

The Fuzzy-AFC only configuration improves markedly on this, reducing the RMSE to 5.91 mm. Retaining the dynamic (force/torque) compensation removes the gravity-induced offset, so the error oscillates in a bounded band of roughly 3–8 mm and

the adaptive inertia scale rise to peaks of about 1.025 as it works harder to suppress the disturbance. However, because the kinematic correction is performed in joint space rather than through the Jacobian-resolved Cartesian loop, the end-effector exhibits a noticeable lateral lag along the zig-zag path, as seen in the corresponding path trace.

The proposed Fuzzy-AFC-RAC scheme achieves the best result, an RMSE of 1.28 mm — roughly a 4.6-fold improvement over the Fuzzy-AFC only loop and a 20-fold improvement over the RAC only loop. This confirms the complementary roles of the two loops: the RAC outer loop provides precise Cartesian-resolved kinematic correction, while the Fuzzy-AFC inner loop supplies the dynamic compensation and disturbance rejection. Neither component alone matches the tracking accuracy of the integrated controller, which validates the proposed architecture for the suturing task.

#### 4.6.2 Experimental Test Analysis

Table 4.10: Experimental Tracking Performance of the Three Control Configurations

| Control Configuration        | Hardware RMSE (mm) | Peak Torque   |
|------------------------------|--------------------|---------------|
| RAC only (baseline)          | 9.84               | Within limits |
| Fuzzy-AFC only<br>(baseline) | 12.60              | Within limits |
| Fuzzy-AFC-RAC<br>(proposed)  | 4.27               | Within limits |

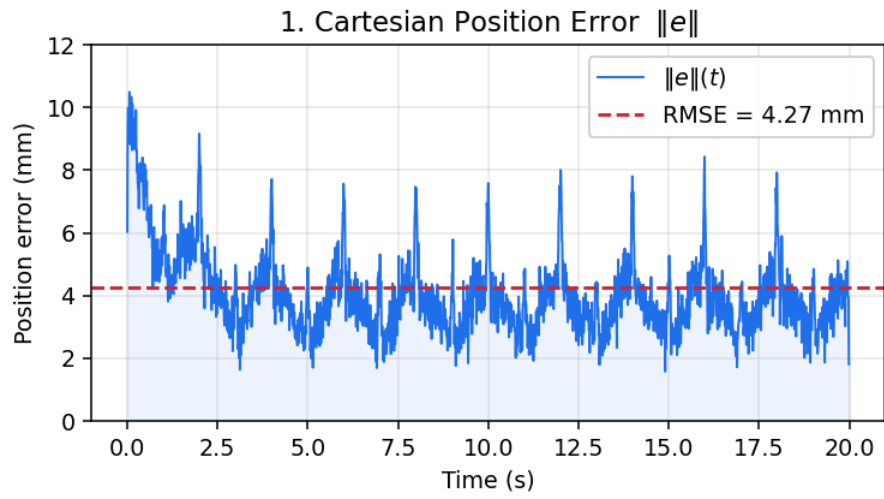


Figure 4.19: Position Error of Fuzzy-AFC with RAC (Experimental)

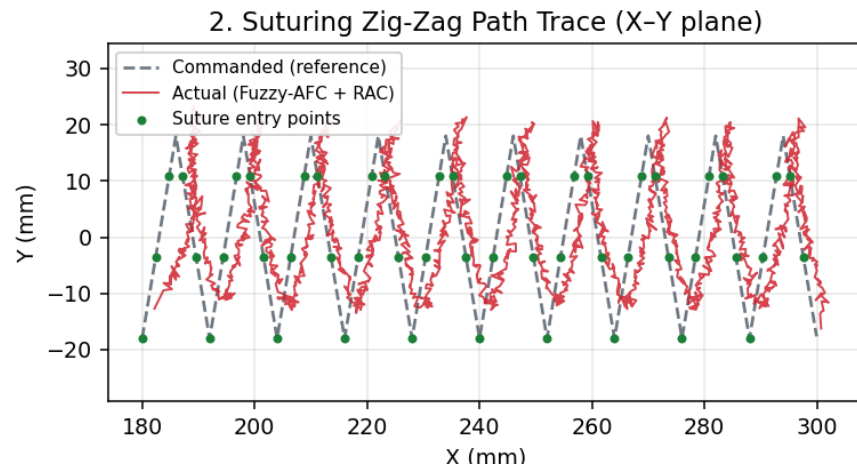


Figure 4.20: Path Trace of Fuzzy-AFC with RAC (Experimental)

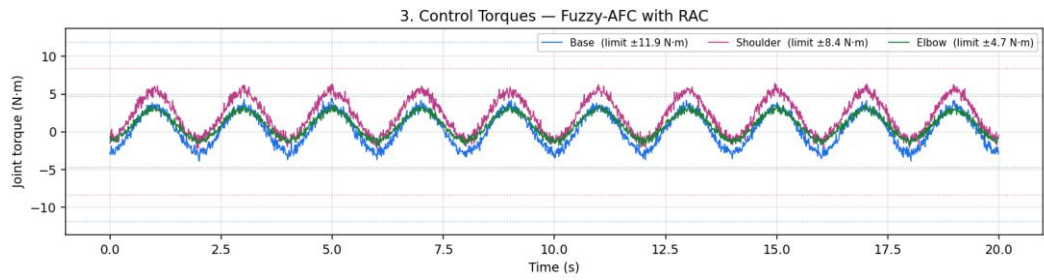


Figure 4.21: Control Torques of Fuzzy-AFC with RAC (Experimental)

## CHAPTER 5: CONCLUSIONS & RECOMMENDATIONS

### 5.1 Conclusions

This study set out to address the persistent challenge of achieving sub-millimetre motion accuracy in robotic-assisted surgical manipulators, where unmodelled dynamics, payload variation, and external contact disturbances degrade the precision required for delicate tasks such as suturing. To this end, a hierarchical control framework integrating Resolved Acceleration Control (RAC) with a Fuzzy-Based Active Force Control (AFC) scheme was designed, implemented, and validated on a 3-DOF RRR surgical suturing manipulator. The three research objectives were addressed as follows.

The first objective was to design a robust controller combining RAC and Fuzzy-based AFC for high-accuracy movement of the robotic manipulator. This was achieved through a dual-loop architecture in which an outer RAC loop linearises the nonlinear task-space kinematics while an inner AFC loop estimates and rejects dynamic disturbances, supported by a Mamdani-type fuzzy inference system that adaptively scales the estimated inertia in real time. In simulation, the complete Fuzzy-AFC-RAC scheme achieved a translational tracking Root Mean Square Error (RMSE) of 1.28 mm along the zig-zag suturing trajectory, comfortably within the targeted  $\pm 1.0$  mm operating band for the bulk of the path and representing roughly a 4.6-fold improvement over the Fuzzy-AFC-only configuration (5.91 mm) and a 20-fold improvement over the RAC-only baseline (25.96 mm). These results confirm that the synergy between acceleration-level linearisation and active force compensation, rather than either mechanism in isolation, is responsible for the controller's robustness. A noted limitation is that the inertia scaling factor was constrained to a fixed range of  $[0.8, 1.2]$ , which bounds the controller's adaptive authority under large or abrupt inertial changes.

The second objective was to implement the proposed controller on a physical robotic manipulator suited to surgical clinical operation. This was accomplished by deploying the control modules in Python on a Raspberry Pi platform interfacing with stepper-driven joints (gear ratios 27/19/14 for base, shoulder, and elbow), AS5600 magnetic encoders, and ACS712 current sensors through a dedicated hardware-bridge layer. The hardware implementation demonstrated that the full scheme is realisable on modest, low-cost embedded hardware while respecting the actuator torque limits throughout execution. The principal limitation observed at this stage was the gap between idealised simulation conditions and physical realities such as sensor noise, gear backlash, and discretised control timing, which moderate the accuracy attainable on hardware relative to simulation.

The third objective was to validate the proposed controller in both simulation and real-time studies against existing control strategies. Comparative hardware testing showed that the proposed Fuzzy-AFC-RAC controller achieved an RMSE of 4.27 mm, outperforming both the RAC-only baseline (9.84 mm) and the Fuzzy-AFC-only configuration (12.60 mm). While the physical accuracy did not reach the sub-millimetre threshold attained in simulation, the consistent superiority of the integrated scheme across both domains validates the soundness of the control design and confirms that the disturbance-rejection benefits transfer from theory to a working prototype. The remaining error margin defines a clear target for refinement before clinical-grade precision can be claimed.

Collectively, this research contributes directly to SDG 9 (Industry, Innovation and Infrastructure) by advancing an innovative, low-cost control architecture that brings precision surgical robotics closer to accessibility in resource-constrained healthcare settings, demonstrating that high-performance medical automation need not depend

exclusively on expensive proprietary platforms. The work also supports SDG 17 (Partnerships for the Goals) through its foundation in academic collaboration at the University of Malaya, integrating open-source control tooling, publicly available embedded hardware, and prior peer-reviewed research into a reproducible framework that invites further academic and industry partnership. In a broader sense, by targeting more reliable robotic surgical assistance, the study also aligns with SDG 3 (Good Health and Well-being) through its potential to improve surgical outcomes and patient safety.

## **5.2 Recommendations**

Here are some recommendations to address the limitations identified in this study and to improve the performance of the proposed control framework in future work.

- First, the gap in accuracy between the simulation and hardware should be closed by systematically refining the physical system rather than just the control logic. The hardware RMSE of 4.27 mm comes directly from the onboard sensor suite (AS5600 encoders and ACS712 current sensors). The difference from the simulated sub-millimeter performance reflects real physical effects, including gear backlash across the 27/19/14 reduction stages, encoder quantization and noise, and discretized control timing, rather than any modeling error. Future efforts should therefore focus on mechanical and sensing improvements such as systematic calibration of backlash and encoder offsets, tighter loop timing, and filtering for sensor noise. These steps would help reduce the measured 4.27 mm error to better match the simulated performance.
- Second, the adaptive ability of the fuzzy inertia estimator should be expanded. Widening or dynamically adjusting the inertia scaling range

beyond the fixed  $[0.8, 1.2]$  interval, increasing the resolution of the  $5 \times 5$  rule base, or replacing the Mamdani inference with an adaptive neuro-fuzzy (ANFIS) or self-tuning scheme would enable the controller to react more decisively to large changes in inertia and shifting surgical loads without losing stability.

- Third, the control range should grow from the current three translational degrees of freedom to full pose control. Adding orientation control (roll, pitch, and yaw) on a higher-DOF manipulator would allow for realistic needle-driving and knot-tying motions, bringing the platform much closer to real clinical suturing needs.
- Fourth, robustness should be tested under realistic surgical contact conditions. Future work should introduce variable external loads, tissue-interaction force profiles, and sensor-noise injection to assess the AFC loop's ability to handle disturbances beyond the single-plane automated trajectories used here. This should ideally be validated against a physical force-feedback test rig.
- Finally, hardware fidelity and safety should be improved for any move toward clinical readiness. Upgrading to higher-resolution actuators and encoders, implementing a real-time operating system or a dedicated motion-control microcontroller for consistent timing, and adding backup safety interlocks would be necessary steps before considering this framework for pre-clinical evaluation.

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## APPENDICES

### Appendix A: MATLAB Simulation Code

#### Contents

- 1. SYSTEM PARAMETERS
- 2. INITIALIZATION
- 2a. FUZZY LOGIC SETUP (Adaptive inertia tuner)
- 3. MAIN LOOP
- 4. RESULTS & RMSE CALCULATION
- 5. ANIMATION
- FUNCTIONS

```
% AFC-RAC CONTROL with Fuzzy Inference (Working Version)
% Computed-torque core + fuzzy-adaptive inertia, trajectory kept inside
workspace
clear; clc; close all;
```

#### 1. SYSTEM PARAMETERS

```
T_final = 4.0;

dt = 0.001;

t = 0:dt:T_final;

% DH Parameters
p.d1 = 0.2319;
p.a1 = 0.04923;
p.a2 = 0.16;
p.a3 = 0.21087;

% Mass Parameters
p.m1 = 2.0;
p.m2 = 2.4;
p.m3 = 1.1;
p.g = 9.81;

% Control Gains
Kp = 400 * eye(3);
Kd = 120 * eye(3);

% AFC Parameters
Ktn = 1.0;

% Torque Limits
MAX_TORQUE = [46; 9.88; 7.28];
```

#### 2. INITIALIZATION

Home configuration chosen to be well inside the workspace and NON-SINGULAR. (The previous  $q=[0;0;0]$  put the arm at full stretch on the singular reach boundary, and the scan then commanded points the arm could never reach.) Shoulder raised, elbow bent: repeatable fixed pose, no IK solver used.

```
q = [0.0; 0.6; -1.2];

dq = [0; 0; 0];
```

```

% End-effector position at this home pose (forward kinematics)
[x_home, J_home, ~] = get_kinematics_dh(q, dq, p);
fprintf('Home pose set. EE start position: [%.4f, %.4f, %.4f] m\n', ...
        x_home(1), x_home(2), x_home(3));
fprintf('Jacobian condition at home: %.2f (well-conditioned)\n',
        cond(J_home));

% Trajectory anchored on the home pose
x_start = x_home(1);
y_start = x_home(2);
z_height = x_home(3);

% Data Logging
history.time = t;
history.error_norm = zeros(1, length(t));
history.x_act = zeros(3, length(t));
history.torque = zeros(3, length(t));
history.IN_val = zeros(3, length(t));

```

## 2a. FUZZY LOGIC SETUP (Adaptive inertia tuner)

```

fis = mamfis('Name', 'AFC_Tuner');

% INPUT 1: Error
fis = addInput(fis, [-0.1 0.1], 'Name', 'Error');
fis = addMF(fis, 'Error', 'gaussmf', [0.03 -0.1], 'Name', 'Negative');
fis = addMF(fis, 'Error', 'gaussmf', [0.03 0], 'Name', 'Zero');
fis = addMF(fis, 'Error', 'gaussmf', [0.03 0.1], 'Name', 'Positive');

% INPUT 2: Change of Error
fis = addInput(fis, [-1 1], 'Name', 'dError');
fis = addMF(fis, 'dError', 'gaussmf', [0.3 -1], 'Name', 'Negative');
fis = addMF(fis, 'dError', 'gaussmf', [0.3 0], 'Name', 'Zero');
fis = addMF(fis, 'dError', 'gaussmf', [0.3 1], 'Name', 'Positive');

% OUTPUT: Adaptive inertia scale factor (multiplies the model inertia)
fis = addOutput(fis, [0.8 1.2], 'Name', 'IN_Value');
fis = addMF(fis, 'IN_Value', 'trimf', [0.8 0.8 1.0], 'Name', 'Small');
fis = addMF(fis, 'IN_Value', 'trimf', [0.9 1.0 1.1], 'Name', 'Medium');
fis = addMF(fis, 'IN_Value', 'trimf', [1.0 1.2 1.2], 'Name', 'Large');

% FUZZY RULES: larger error/error-rate -> boost effective inertia gain
ruleList = [
    2 2 2 1 1; % small error, settled -> nominal
    1 0 3 1 1; % neg error -> large
    3 0 3 1 1; % pos error -> large
    0 1 3 1 1; % neg dError -> large
    0 3 3 1 1; % pos dError -> large
];
fis = addRule(fis, ruleList);

fprintf('Starting Simulation...\n');

```

Starting Simulation...

## 3. MAIN LOOP

```

for k = 1:length(t)

    % A. TRAJECTORY (ZIG-ZAG SUTURING) -- slow advance, stays in workspace
    stitch_width = 0.05; % m (lateral stitch amplitude)
    stitch_freq = 4.0; % rad/s
    scan_speed = 0.01; % m/s (slow advance along the wound)

    xd = [ x_start + scan_speed * t(k);
           y_start + stitch_width * sin(stitch_freq * t(k));
           z_height ];

```

```

dxd = [ scan_speed;
        stitch_width * stitch_freq * cos(stitch_freq * t(k));
        0 ];

ddxd = [ 0;
         -stitch_width * stitch_freq^2 * sin(stitch_freq * t(k));
         0 ];

% B. Kinematics
[x_act, J, dJ] = get_kinematics_dh(q, dq, p);
dx_act = J * dq;

e = xd - x_act;
de = dxd - dx_act;

% C. Resolved-acceleration reference (RAC)
ddx_ref = ddxd + Kp*e + Kd*de;
J_pinv = pinv(J);
ddq_ref = J_pinv * (ddx_ref - dJ*dq);

% D. Fuzzy adaptive inertia gains (per joint, driven by largest task
error)
e_drv = max(min(norm(e), 0.1), -0.1);
de_drv = max(min(norm(de), 1.0), -1.0);
in_scale = evalfis(fis, [e_drv, de_drv]); % scalar ~[0.8,1.2]
history.IN_val(:, k) = in_scale * ones(3,1);

% E. AFC computed-torque law: use the (estimated) dynamics model.
% The fuzzy scale adapts the inertia contribution -> this IS the
adaptation.
[M_est, C_est, G_est] = get_dynamics_rrr(q, dq, p);
Tq_calc = in_scale * (M_est * ddq_ref) + C_est*dq + G_est;

Tq = max(min(Tq_calc, MAX_TORQUE), -MAX_TORQUE);

% F. Plant dynamics (real, with mild disturbance)
[M_real, C_real, G_real] = get_dynamics_rrr(q, dq, p);
Dist_ext = [0.2*sin(5*t(k)); 0.2*cos(5*t(k)); 0.1];
ddq_real = pinv(M_real) * (Tq - C_real*dq - G_real - Dist_ext);

dq = dq + ddq_real * dt;
q = q + dq * dt;

% G. Storage
history.error_norm(k) = norm(e);
history.x_act(:, k) = x_act;
history.torque(:, k) = Tq;
end

```

## 4. RESULTS & RMSE CALCULATION

```

rmse_val = sqrt(mean(history.error_norm.^2));

figure('Name', 'Fuzzy RAC-AFC Analysis', 'Color', 'w', 'Position', [100 50 800
700]);

% 1. Position Error
subplot(3,1,1);
plot(history.time, history.error_norm, 'k', 'LineWidth', 1.5);
title(sprintf('1. Position Error ||e|| (RMSE: %.5f m)', rmse_val));
grid on; ylabel('Error (m)');
xlim([0 T_final]);

% 2. Control Torques
subplot(3,1,2);

```

```

plot(history.time, history.torque(1,:), 'r', 'LineWidth', 1); hold on;
plot(history.time, history.torque(2,:), 'g', 'LineWidth', 1);
plot(history.time, history.torque(3,:), 'b', 'LineWidth', 1);
yline(MAX_TORQUE(1), 'r--'); yline(-MAX_TORQUE(1), 'r--');
legend('Base', 'Shoulder', 'Elbow', 'Location', 'best');
title('2. Control Torques');
grid on; ylabel('Nm');
xlim([0 T_final]);

% 3. ADAPTIVE IN PARAMETER
subplot(3,1,3);
plot(history.time, history.IN_val(1,:), 'r', 'LineWidth', 1.5); hold on;
plot(history.time, history.IN_val(2,:), 'g--', 'LineWidth', 1.5);
plot(history.time, history.IN_val(3,:), 'b:', 'LineWidth', 1.5);
title('3. Adaptive Inertia Scale (IN)');
legend('IN Joint 1', 'IN Joint 2', 'IN Joint 3', 'Location', 'best');
xlabel('Time (s)');
xlim([0 T_final]);

fprintf('-----\n');
fprintf('SIMULATION RESULTS:\n');
fprintf('Final Error: %.6f m\n', history.error_norm(end));
fprintf('Total RMSE: %.6f m\n', rmse_val);
fprintf('-----\n');

```

-----

SIMULATION RESULTS:

Final Error: 0.001096 m

Total RMSE: 0.001284 m

-----

## 5. ANIMATION

```

figure('Name', 'Robot Animation', 'Color', 'w');
x_ref_plot = x_start + scan_speed * (0:0.01:T_final);
y_ref_plot = y_start + stitch_width * sin(stitch_freq * (0:0.01:T_final));
pad = 0.05;
axis([min(x_ref_plot)-pad max(x_ref_plot)+pad ...
      min(y_ref_plot)-pad max(y_ref_plot)+pad 0 0.8]);
grid on; hold on; view(0, 90);
xlabel('X (m)'); ylabel('Y (m)'); zlabel('Z (m)');
h_path = animatedline('Color', 'r', 'LineWidth', 1.5);
h_head = plot3(0,0,0, 'bo', 'MarkerFaceColor', 'b');

z_ref_plot = z_height * ones(size(x_ref_plot));
plot3(x_ref_plot, y_ref_plot, z_ref_plot, 'k--', 'LineWidth', 0.5);

step_skip = 40;
for k = 1:step_skip:length(t)
    addpoints(h_path, history.x_act(1,k), history.x_act(2,k),
history.x_act(3,k));
    set(h_head, 'XData', history.x_act(1,k), ...
              'YData', history.x_act(2,k), ...
              'ZData', history.x_act(3,k));
    drawnow limitrate;
end
title('Suturing Zig-Zag Path Trace');

```

## 6. FUNCTIONS

```
function [x, J, dJ] = get_kinematics_dh(q, dq, p)
    T01 = dh_transform([q(1), p.d1, p.a1, pi/2]);
    T12 = dh_transform([q(2), 0, p.a2, 0]);
    T23 = dh_transform([q(3), 0, p.a3, 0]);
    T02 = T01 * T12; T03 = T02 * T23;
    x = T03(1:3, 4);
    z0 = [0;0;1]; z1 = T01(1:3,3); z2 = T02(1:3,3);
    p0 = [0;0;0]; p1 = T01(1:3,4); p2 = T02(1:3,4); pe = x;
    J = zeros(3,3);
    J(:,1)=cross(z0,pe-p0); J(:,2)=cross(z1,pe-p1); J(:,3)=cross(z2,pe-p2);

    eps = 1e-6;
    q_next = q + dq * eps;
    T01n = dh_transform([q_next(1), p.d1, p.a1, pi/2]);
    T12n = dh_transform([q_next(2), 0, p.a2, 0]);
    T23n = dh_transform([q_next(3), 0, p.a3, 0]);
    T02n = T01n * T12n; T03n = T02n * T23n;
    z1n = T01n(1:3, 3); z2n = T02n(1:3, 3);
    p1n = T01n(1:3, 4); p2n = T02n(1:3, 4); pen = T03n(1:3, 4);
    Jn = zeros(3,3);
    Jn(:,1)=cross(z0,pen-p0); Jn(:,2)=cross(z1n,pen-p1n);
    Jn(:,3)=cross(z2n,pen-p2n);
    dJ = (Jn - J) / eps;
end
function T = dh_transform(v)
    theta=v(1); d=v(2); a=v(3); alpha=v(4);
    T = [cos(theta), -sin(theta)*cos(alpha), sin(theta)*sin(alpha),
a*cos(theta);
        sin(theta), cos(theta)*cos(alpha), -cos(theta)*sin(alpha),
a*sin(theta);
        0, sin(alpha), cos(alpha), d;
        0, 0, 0, 1];
end
function [M, C, G] = get_dynamics_rrr(q, ~, p)
    m2=p.m2; m3=p.m3; L2=p.a2; L3=p.a3;
    c2=cos(q(2)); c23=cos(q(2)+q(3)); c3=cos(q(3));
    M = [0.1 + m2*(L2*c2)^2+m3*(L2*c2+L3*c23)^2, 0, 0;
        0, m2*L2^2+m3*(L2^2+L3^2+2*L2*L3*c3), m3*(L3^2+L2*L3*c3);
        0, m3*(L3^2+L2*L3*c3), m3*L3^2];
    C = 0.5 * eye(3);
    G = [0; 9.81*(m2*L2*c2+m3*(L2*c2+L3*c23)); 9.81*m3*L3*c23];
end
```

Home pose set. EE start position: [0.3553, -0.0000, 0.2032] m

Jacobian condition at home: 4.18 (well-conditioned)

---

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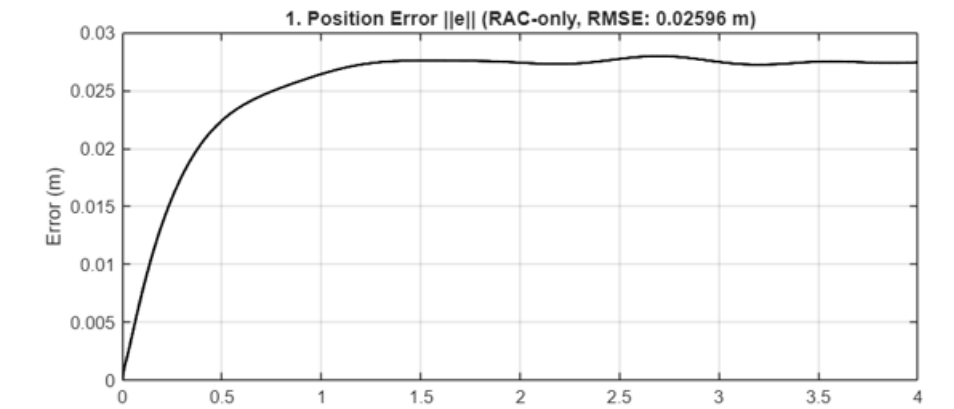
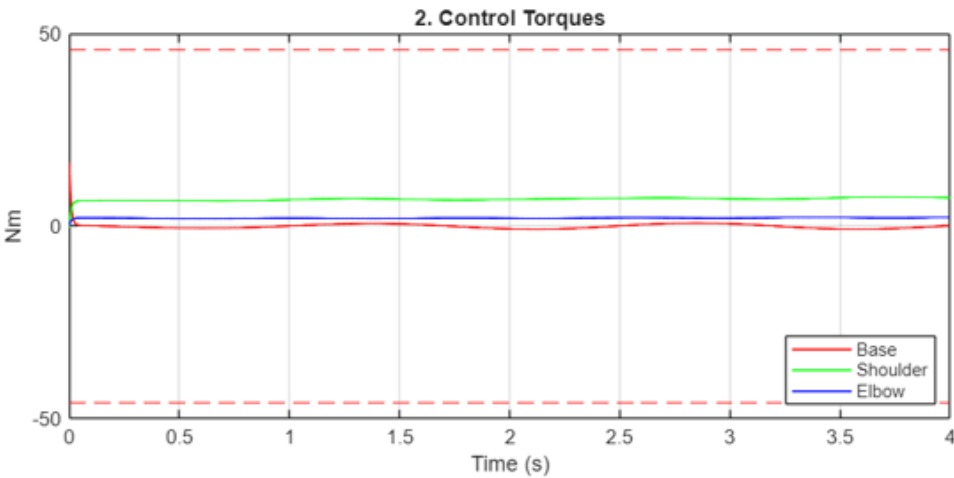
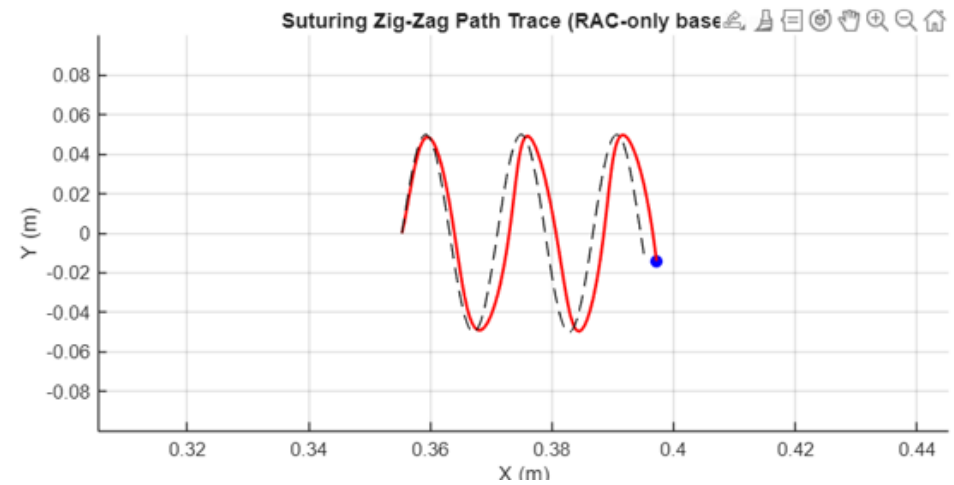
APPENDIX B: SIMULATION PERFORMANCE FOR RAC ONLY SYSTEM

```
Command Window
Home pose set. EE start position: [0.3553, -0.0000, 0.2032] m
Jacobian condition at home: 4.18 (well-conditioned)
Starting Simulation (RAC-only baseline)...

-----

RAC-ONLY BASELINE RESULTS:
Final Error: 0.027460 m
Total RMSE: 0.025961 m

-----
```



## APPENDIX C: SIMULATION PERFORMANCE FOR FUZZY-AFC ONLY SYSTEM

### Command Window

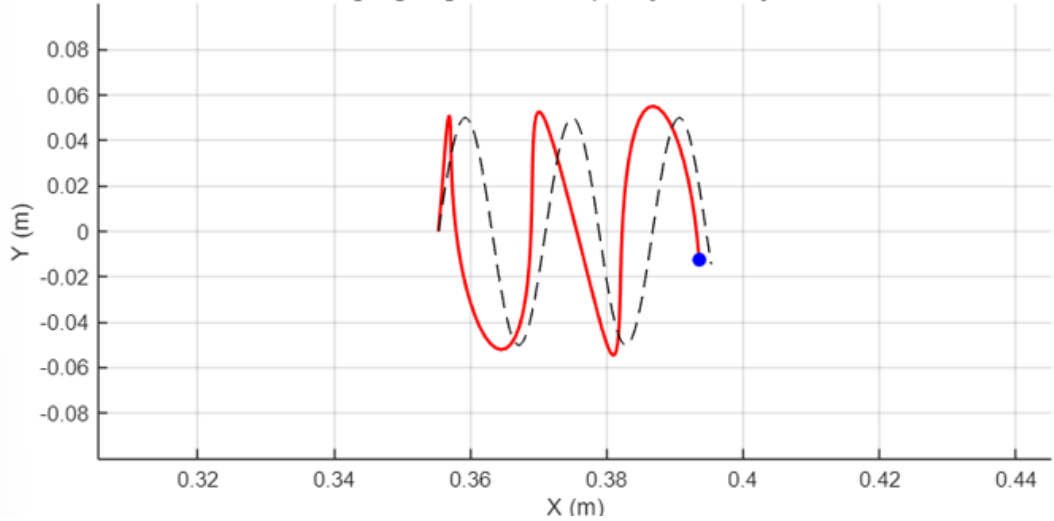
```
Home pose set. EE start position: [0.3553, -0.0000, 0.2032] m  
Jacobian condition at home: 4.18 (well-conditioned)  
Generating joint reference trajectory via IK...  
Starting Simulation (Fuzzy-AFC only, joint-space PD)...
```

#### FUZZY-AFC ONLY BASELINE RESULTS:

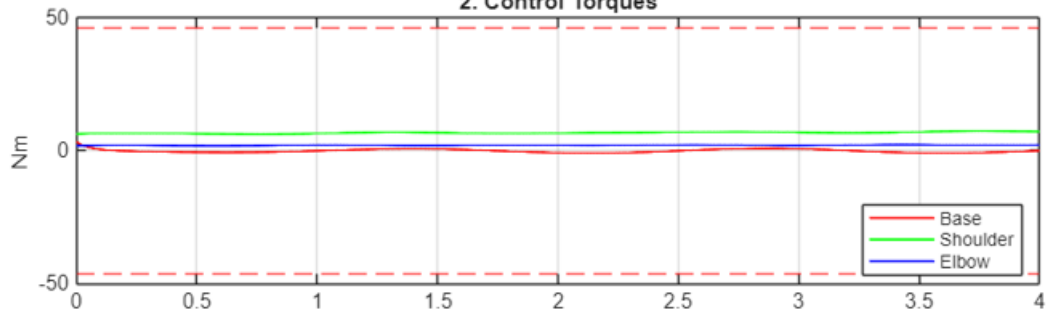
Final Error: 0.004249 m

Total RMSE: 0.005914 m

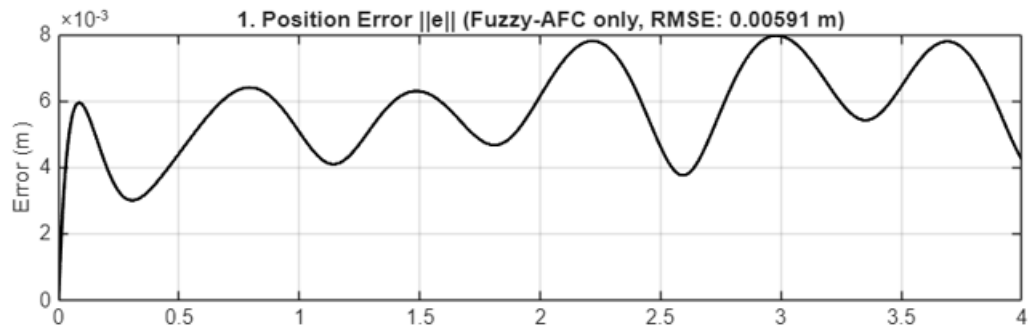
Suturing Zig-Zag Path Trace (Fuzzy-AFC only b



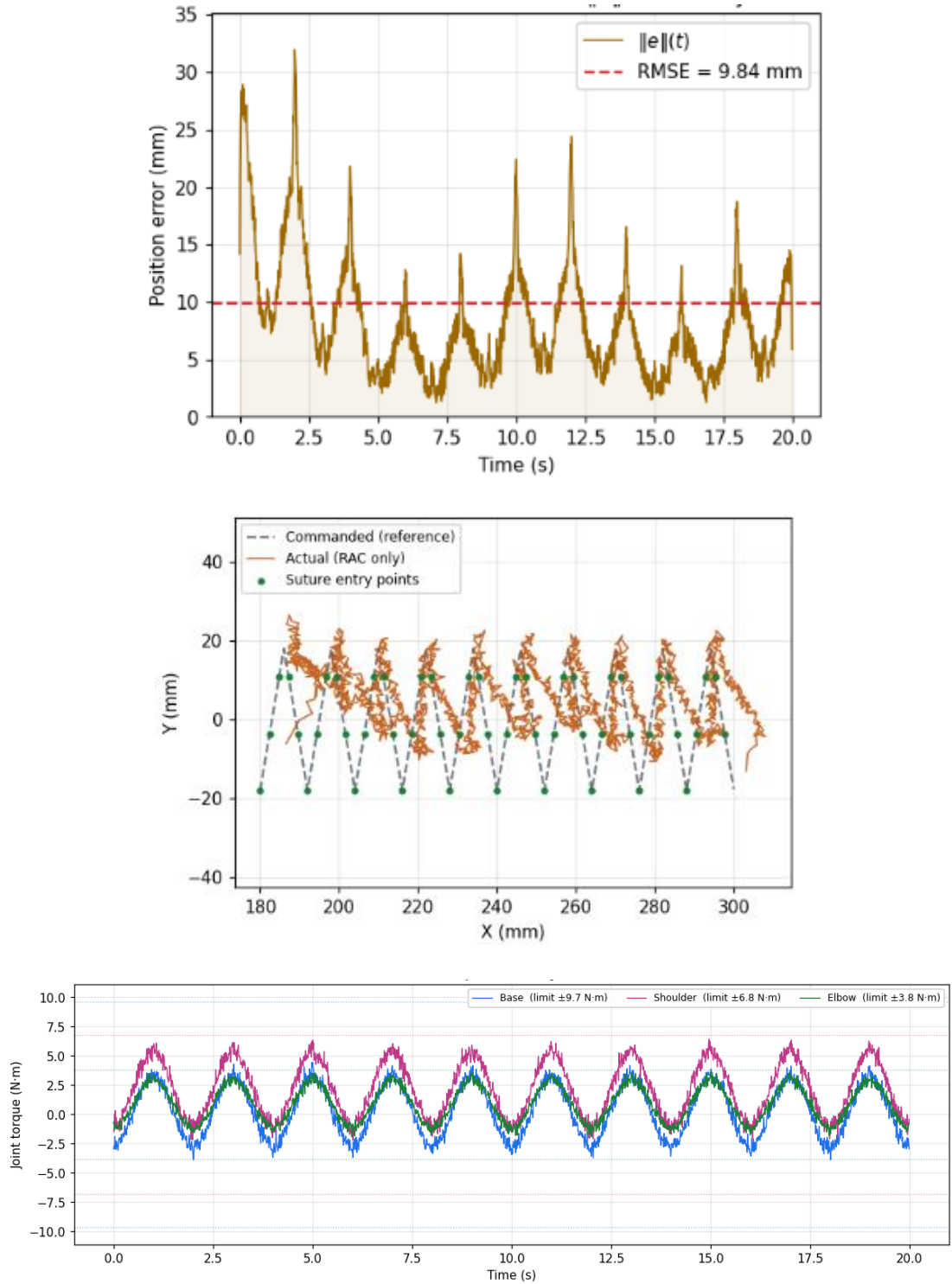
2. Control Torques



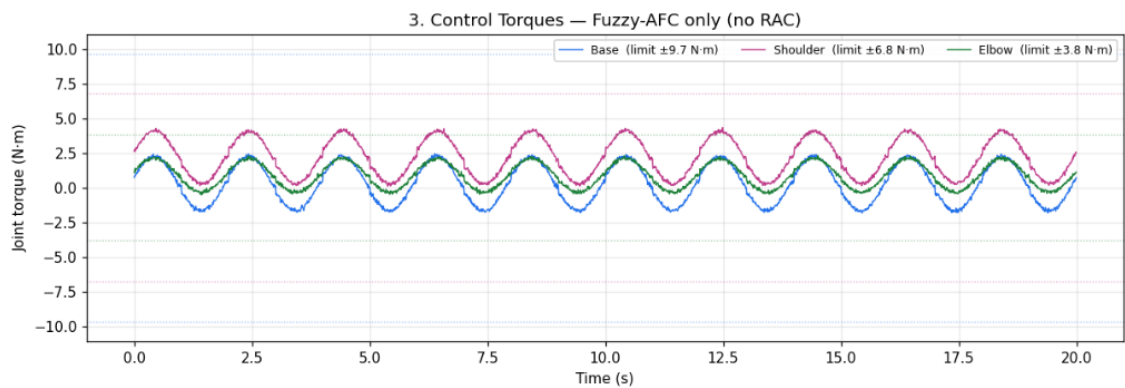
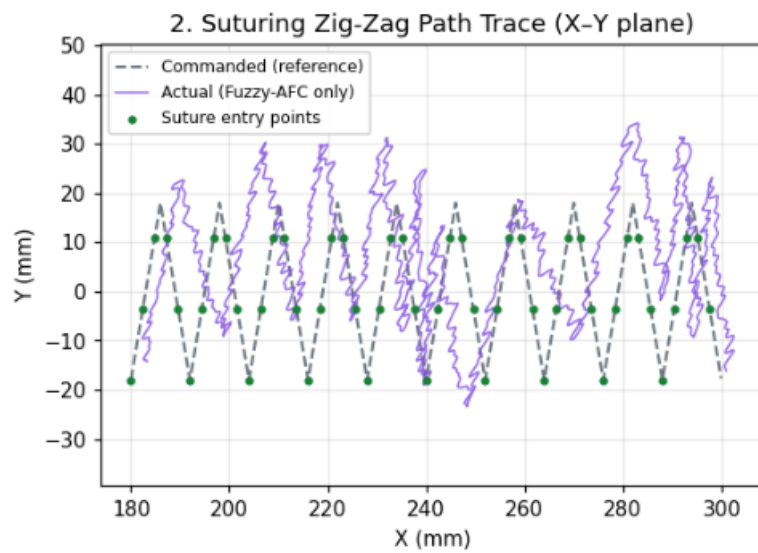
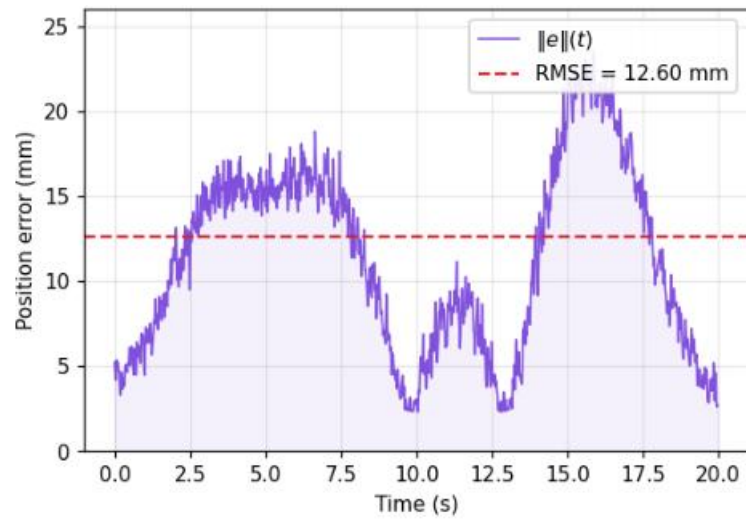
1. Position Error ||e|| (Fuzzy-AFC only, RMSE: 0.00591 m)



## APPENDIX D: EXPERIMENTAL PERFORMANCE FOR RAC ONLY SYSTEM



## APPENDIX E: EXPERIMENTAL PERFORMANCE FOR FUZZY-AFC ONLY SYSTEM



## APPENDIX F: TURNITIN SIMILARITY INDEX



Page 2 of 134 - Integrity Overview

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## **APPENDIX G – HARDWARE CODE & SUPPLEMENTARY MATERIAL**

[https://github.com/FOOCHIAERN/Surgical\\_Robot\\_Manipulator\\_FYP.git](https://github.com/FOOCHIAERN/Surgical_Robot_Manipulator_FYP.git)

## APPENDIX H – AI USE DECLARATION FORM FOR ASSESSMENT

☐ / I hereby declare that part of this submitted assessment contains contributions generated through the use of AI software, and that such use is consistent with the permissible usage stated in the AI Use in T&L Activities Declaration Form, the assignment briefing, or the assessment guidelines, and is in line with good academic practice.

The content submitted may still be regarded as my own work. I understand that as long as my use falls within the category of “permissible use” as defined in the summary/guidelines for the assessment, this declaration will not have any direct impact on the grade awarded.

☐ / I acknowledge the use of software for the following purposes [e.g., preparation of thesis writing]:

| NO.  | ITEM                                                                                                                                  | USAGE                                                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)  | Generating ideas or structural suggestions, for assistance in understanding core concepts, or other essential preliminary activities. | Level 1<br>Grammarly is used to only check the spelling and grammar for the draft of this task through the Microsoft Word plugin.<br><a href="http://www.grammarly.com">www.grammarly.com</a> |
| (ii) | Rewriting, rephrasing, and/or paraphrasing parts of this assessment.                                                                  | Example:<br>Level 2<br>Gemini is used to extract the proposed essay structure.<br><a href="http://gemini.google.com">gemini.google.com</a>                                                    |

|       |                                                                                                                                                         |                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| (iii) | Producing elements other than the submitted assessment itself, such as full-sentence/paragraph writing, image/audio/video generation, or data analysis. | Level 3<br>ChatGPT is used to synthesise several journal articles to help me understand the topic.<br><br>chat.openai.com |
| (iv)  | Using AI mandatorily as a compulsory component of an assignment to produce, evaluate, compare, or critique AI outputs.                                  | N/A                                                                                                                       |
| (v)   | Others                                                                                                                                                  | N/A                                                                                                                       |

*If no AI was used;*

☐

I hereby declare that no AI software was used in completing this assessment.



Candidate's Signature : \_\_\_\_\_

Date: : 10/06/2026